

Brønsted—Lowry acids and bases

Learning outcomes

By the end of this section you should be able to deduce the Brønsted—Lowry acid and base in a reaction.

Acidic substances give away a proton to other substances that can accept them. A hydrogen atom that has lost an electron is a proton.

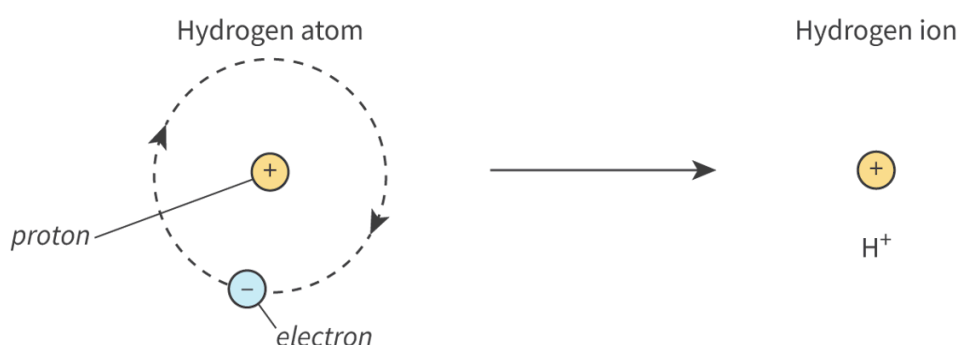


Figure 1. A hydrogen ion is a proton.

 More information for figure 1

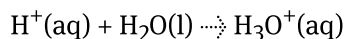
The diagram illustrates the transformation of a hydrogen atom into a hydrogen ion. On the left, it depicts a hydrogen atom consisting of a proton labeled with a positive sign in the center, encircled by a dashed line, representing an electron orbit with a negatively labeled electron. On the right side, an arrow leads to a simplified representation of a hydrogen ion, shown as just a proton with a positive charge symbol (H^+). The diagram indicates the loss of the electron, resulting in the formation of a hydrogen ion.

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The Brønsted–Lowry theory defines acids and bases as either proton donors or proton acceptors. According to the theory:

- a Brønsted–Lowry acid donates a proton (is a proton donor)
- a Brønsted–Lowry base accepts a proton (is a proton acceptor)

A proton refers to a hydrogen ion H^+ which exists as the hydronium ion H_3O^+ in aqueous solution.



Study skills

A hydrogen ion in aqueous solution can be represented as $\text{H}^+(\text{aq})$ or $\text{H}_3\text{O}^+(\text{aq})$.

Nature of Science

Aspect: Theories

Why has the definition of acid evolved over time?

The central observable properties of acids and bases have been known for many hundreds of years. Acid solutions have a sour taste and react with a range of substances such as metal oxides and carbonates. Alkali solutions have a soapy feel and a bitter taste.

Lavoisier was the first to attempt to give a chemical rationale to the common properties of acids by suggesting that they all contained oxygen. It is true that many common acids are indeed oxyacids, but this idea was negated by the identification of hydrocyanic acid, HCN, and hydrochloric acid, HCl.

The German chemist, Justus von Liebig, first put forward the idea that all acids contained hydrogen in 1830, and also that this hydrogen could be replaced by metals in reactions.

Subsequent studies showed that acids, alkalis and salts could be grouped together as electrolytes — their solutions all conducted electricity and contained ions. The observation of a range of acids and alkalis, together with the products from their reactions, led to an explanation of their nature in terms of the ions present in their solutions.

Spectroscopic evidence has demonstrated that it is the hydronium ion, H_3O^+ (a protonated water molecule), which is the active ion present in acidic solutions (Brønsted–Lowry theory). These studies give substance to the ionic theory of acid–base behaviour.

A lot of reactions in life involve the transfer of protons from substances called acids to substances called bases. Did you know, for example, that one of these reactions is used by geologists to prove that a mineral is made of calcite? Watch **Video 1** to see how.

How to Test a Mineral with HCl, the "Acid Test" to Identify Ca...



Video 1. The test for calcite.

The bubbling is the direct result of a proton, or a hydrogen cation, being transferred from the acid to the calcium carbonate producing carbon dioxide gas, water and calcium chloride. Can you relate this reaction to the one happening when acid rain falls on material that is made of limestone or calcium carbonate as seen in subtopic S3.1 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43116/>)? Indeed, this same transfer of protons also occurs in the clouds, when water reacts with acidic fumes from the combustion of fossil fuels as seen in subtopic R1.3 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43475/>). Then this is also what happens when the resulting acidic rain reacts with marble statues, or when an increasingly acidic ocean reacts with coral reefs, or when you pour vinegar on baking powder to produce ‘volcanoes’.

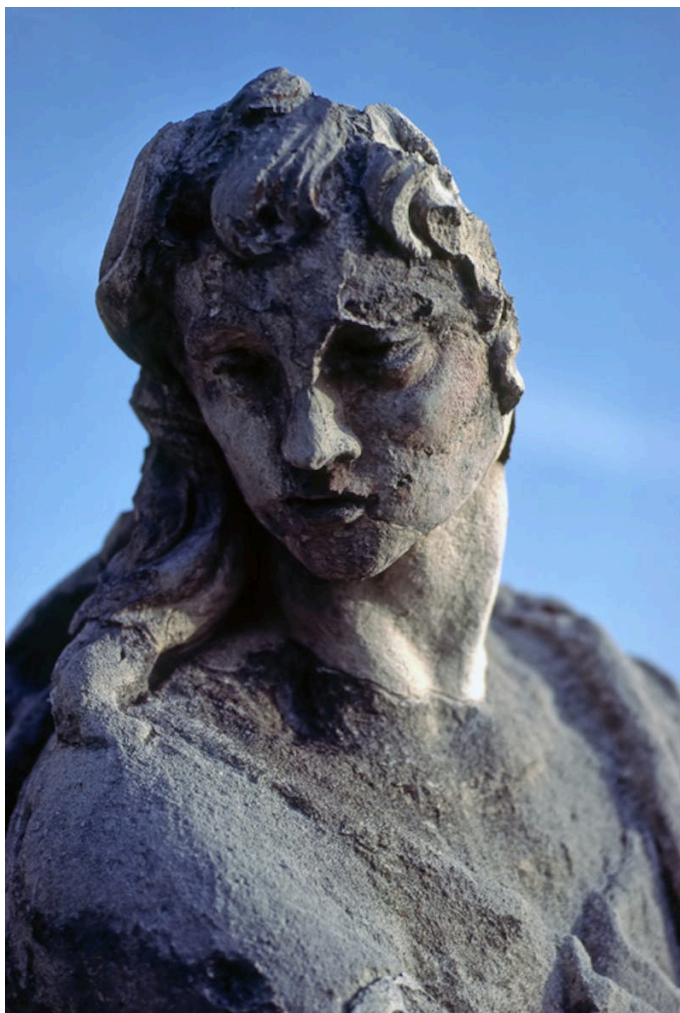


Figure 2. The effect of acid rain on monuments.

Credit: julof90, Getty Images

When hydrogen chloride gas dissolves in water, it donates a proton (H^+) to the water molecule, which accepts the proton.

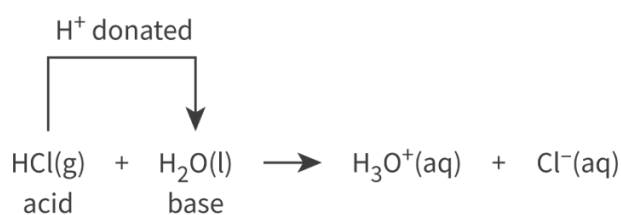


Figure 3. The reaction between hydrogen chloride gas and water.

 More information for figure 3

The image depicts a chemical diagram illustrating the reaction between hydrogen chloride gas (HCl) and water (H_2O) to form hydronium ions (H_3O^+) and chloride ions (Cl^-). The diagram includes labeled components. HCl(g) , labeled as 'acid', and $\text{H}_2\text{O(l)}$, labeled as 'base', are the reactants shown on the left. An arrow labeled ' H^+ donated' points from above HCl(g) to $\text{H}_2\text{O(l)}$, indicating the donation of a proton. To the right of the arrow indicating the direction of the reaction, the products are shown as $\text{H}_3\text{O}^+(\text{aq})$

and $\text{Cl}(\text{aq})$. This visual representation illustrates the Brønsted—Lowry acid-base reaction concept, where HCl acts as an acid by donating a proton to water, forming a hydronium ion.

[Generated by AI]

The hydrogen chloride gas is acting as a Brønsted–Lowry acid and the water molecule as a Brønsted–Lowry base. The solution formed is named hydrochloric acid and is an example of a Brønsted–Lowry acid.

Study skills

All Brønsted—Lowry acids contain hydrogen, as they must be able to donate a hydrogen ion to another species.

Substances that can accept a proton are called Brønsted–Lowry bases. Ammonia (NH_3) is an example of a Brønsted–Lowry base. When ammonia reacts with water, it accepts a proton from the water molecule which acts as a Brønsted–Lowry acid.

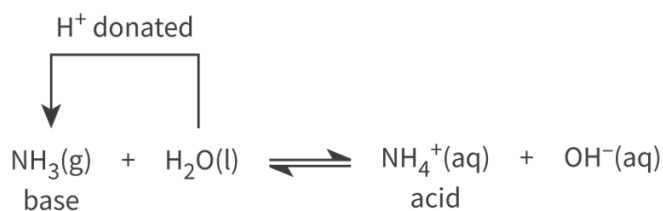


Figure 4. The reaction of ammonia and water.

 More information for figure 4

The image is a diagram illustrating the chemical reaction between ammonia (NH_3) and water (H_2O). It shows ammonia acting as a Brønsted—Lowry base by accepting a proton (H^+) from water. The left side of the equation shows NH_3 in gaseous form and H_2O in liquid form. An arrow labeled "H⁺ donated" indicates the transfer of a proton from H_2O to NH_3 . This results in the formation of NH_4^+ (ammonium ion) and OH^- (hydroxide ion) on the right side of the equation. NH_4^+ is labeled as an acid, showing its potential to donate a proton in subsequent reactions, while OH^- remains as the conjugate base. The chemical equilibrium is represented by a double arrow between the reactants and products.

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Study skills

All Brønsted–Lowry bases must have a lone pair of electrons to form a bond with the hydrogen ion donated to them by the Brønsted–Lowry acid.

Names and formulae of common acids

In this DP chemistry course you will learn about three types of acid:

- binary acids
- oxyacids
- organic acids

Binary acids (see [section S2.1.2 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ionic-bonding-id-43557/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ionic-bonding-id-43557/)) are formed when a halide reacts with hydrogen.

Table 1. Binary acids

Name of acid (IUPAC name)	Formula
Hydrofluoric acid (aqueous hydrogen fluoride)	HF(aq)
Hydrochloric acid (aqueous hydrogen chloride)	HCl(aq)
Hydrobromic acid (aqueous hydrogen bromide)	HBr(aq)
Hydroiodic acid (aqueous hydrogen iodide)	HI(aq)

Oxyacids (see [subtopic S2.1 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43108/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43108/)) are derived from polyatomic ions.

Table 2. Oxyacids acids.

Name of acid	Formula
Nitric acid (aqueous hydrogen nitrate)	$\text{HNO}_3(\text{aq})$
Sulfuric acid (aqueous hydrogen sulfate)	$\text{H}_2\text{SO}_4(\text{aq})$
Carbonic acid (aqueous hydrogen carbonate)	$\text{H}_2\text{CO}_3(\text{aq})$
Phosphoric acid (aqueous hydrogen phosphate)	$\text{H}_3\text{PO}_4(\text{aq})$

Organic acids (see [subtopic S3.2 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43122/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43122/)) contain a carboxyl group.

Table 3. Organic acids.

Name of acid	Formula
Ethanoic acid	$\text{CH}_3\text{COOH}(\text{aq})$
Benzoic acid	$\text{C}_6\text{H}_5\text{COOH}(\text{s})$

Although acids donate a hydrogen ion, not all substances that contain hydrogen behave as acids. For example, methane (CH_4) contains four hydrogen atoms, but does not donate any of them. Therefore, methane does not behave as an acid.

Acids can be classified as monoprotic, diprotic or triprotic depending on the number of hydrogen ions or protons that they can donate in an acid-base reaction.

Hydrochloric acid and nitric acid are monoprotic acids because they can only donate one hydrogen ion in acid-base reactions. Note that ethanoic acid is also monoprotic despite having four hydrogen atoms as only one of these is donated when ethanoic acid reacts with a base such as ammonia, NH_3 .

Sulfuric acid is an example of a diprotic acid because it can donate two hydrogen ions in acid-base reactions.

Phosphoric acid is an example of a triprotic acid because it can donate three hydrogen ions in acid-base reactions.

Names and formulae of common bases

Many different substances are thought of as being bases. Some of the most common are discussed below.

Metal hydroxides and oxides (see [section R3.1.7](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/reactions-of-acids-id-46056/>))) that combine with water are bases. The names and formulae of some common bases are shown in **Table 4**.

Table 4. Oxides and hydroxide bases.

Name of base	Formula
Calcium oxide	CaO(s)
Copper(II) oxide	CuO(s)
Sodium hydroxide	NaOH(aq)
Lithium hydroxide	LiOH(aq)
Calcium hydroxide	Ca(OH) ₂ (aq)
Barium hydroxide	Ba(OH) ₂ (aq)

Carbonates and metal hydrogencarbonates are also examples of weak bases (see [section R3.1.7](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/reactions-of-acids-id-46056/>))). Some examples are shown in **Table 5**.

Table 5. Carbonates and hydrogen carbonates.

Name of base	Formula
Sodium hydrogen carbonate	NaHCO ₃ (aq)
Sodium carbonate	Na ₂ CO ₃ (aq)

Ammonia, amines and amides are also examples of weak bases (see [section R3.1.7](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/reactions-of-acids-id-46056/>))).

Table 6. Ammonia, amines and amides.

Name of base	Formula
Ammonia	$\text{NH}_3(\text{aq})$
Ethanamine	$\text{CH}_3\text{CH}_2\text{NH}_2(\text{aq})$

Study skills

Many bases are insoluble — they do not dissolve in water. However, if a base does dissolve in water, it is known as an alkali.

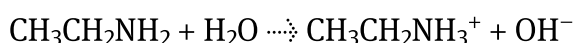
Examples of alkalis include the group 1 metal hydroxides sodium hydroxide and lithium hydroxide. Examples of insoluble bases include the metal oxides calcium oxide and copper(II) oxide.

Worked example 1

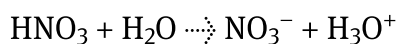
Write the equation for the reaction of:

1. Ethanamine with water
2. Nitric acid with water

1. Ethanamine is a base, so it accepts a proton from water:



2. Nitric acid is an acid, it donates a proton to water:



Activity

- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:**
 - Thinking skills — Being curious about the natural world; Applying key ideas and facts in new contexts
 - Social skills — Actively seeking and considering the perspective of others

- **Time required to complete activity:** 20 minutes
- **Activity type:** Individual activity

Instructions:

1. Go on a hunt around your house and find at least five acids and two bases.
2. In a spreadsheet, list them in the first column of a table with three columns. State their common name and where they are found.
3. In the second column, add their scientific names.
4. In the third column, add their formulas.

Conjugate acids-bases

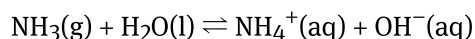
Learning outcomes

By the end of this section you should be able to:

- (R3.1.2) Deduce the formula of the conjugate acid or base of any Brønsted–Lowry acid or base.
- (R3.1.3) Interpret and formulate equations to show acid–base reactions of amphoteric species.

Conjugate acid–base pairs

The reaction between ammonia and water to produce the ammonium ion and the hydroxide ion is:



In the forward reaction, water acts as a Brønsted–Lowry acid and ammonia as a Brønsted–Lowry base. An important point to note is that this is a reversible reaction, indicated by the two-way arrow ($\rightleftharpoons$). This means that the reaction occurs in both directions with the forward reaction occurring at the same rate as the reverse reaction. In other words, the reaction is at equilibrium ([section R2.3.1 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/physical-equilibrium-id-43483/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/physical-equilibrium-id-43483/)). In the reverse reaction, the NH_4^+ ion donates a proton to the OH^- ion (**Figure 1**), which means that the NH_4^+ ion is acting as a Brønsted–Lowry acid and the OH^- ion is acting as a Brønsted–Lowry base.

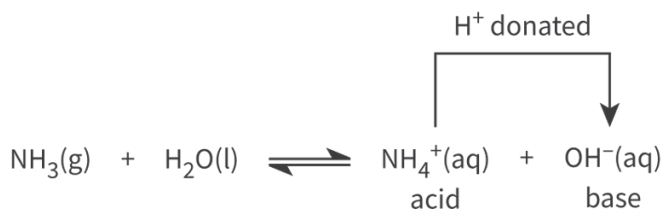


Figure 1. In the reverse reaction, the ammonium ion donates a proton to the hydroxide ion.

📖 More information for figure 1

If we look at NH_3 and NH_4^+ , we can see that they differ by one proton (H^+), therefore they are referred to as a conjugate (acid–base) pair. The H_2O and OH^- ion also differ by a proton and are likewise known as a conjugate acid–base pair

(Figure 2).

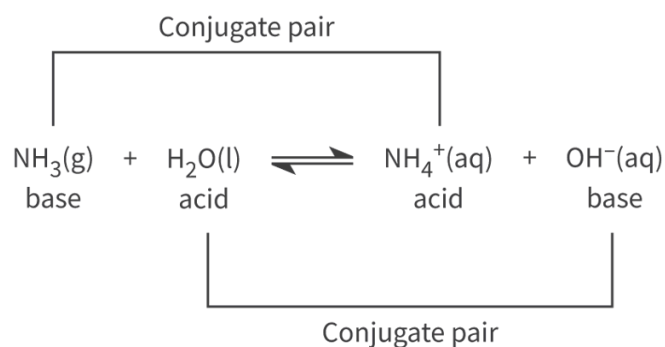


Figure 2. The two conjugate acid—base pairs in the reaction of ammonia and water.

📄 More information for figure 2

Therefore, in the forward reaction:

- The NH_4^+ ion is the conjugate acid of the base NH_3 .
- The OH^- ion is the conjugate base of the acid H_2O .

In the reverse reaction:

- NH_3 is the conjugate base of the acid NH_4^+ .
- H_2O is the conjugate acid of the base OH^- .

Figure 3 shows the reaction between hydrogen chloride gas and water. The hydrogen chloride gas is acting as a Brønsted–Lowry acid and the water is acting as a Brønsted–Lowry base. The two conjugate pairs are HCl and Cl^- and H_2O and H_3O^+ . Once again, we can see the two species in each conjugate pair differ by a single proton (H^+).

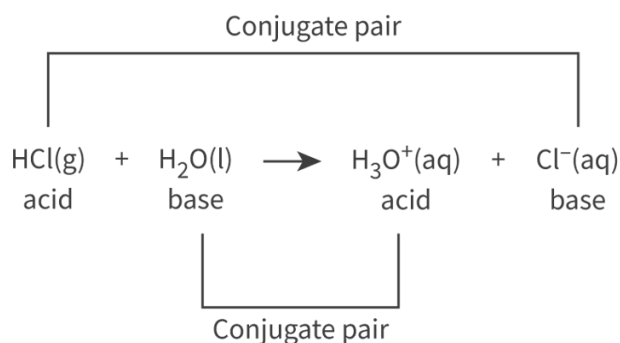


Figure 3. The two conjugate pairs in the reaction of hydrogen chloride gas and water.

 More information for figure 3

Study skills

When writing the formula of the conjugate acid of a base, add one proton (H^+).

Conversely, when writing the conjugate base of an acid, remove a proton (H^+).

The charge on the species must be changed, as protons have a positive charge.

The conjugate base of an acid has one less positive charge, and the conjugate acid of a base has one more positive charge.

Use **Interactive 1** to give the conjugate base of each acid.

Check your understanding! Can you give the conjugate base of the following acids?



 Turn

Card 1 of 5

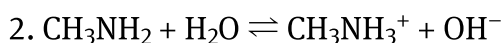
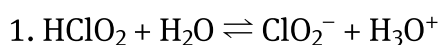


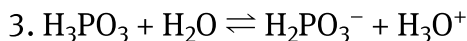
Interactive 1. Find the Conjugate Base of Each Acid.

 More information for interactive 1

Worked example 1

Identify the conjugate acid–base pairs in the reaction equations below:





1. HClO_2 is the acid and ClO_2^- is the conjugate base.
 H_2O is the base and H_3O^+ is the conjugate acid.
2. CH_3NH_2 is the base and CH_3NH_3^+ is the conjugate acid.
 H_2O is the acid and OH^- is the conjugate base.
3. H_3PO_3 is the acid and H_2PO_3^- is the conjugate base.
 H_2O is the base and H_3O^+ is the conjugate acid.

Making connections

We have seen that metal oxides are basic. What is the periodic trend in the acid–base properties of metal and non-metal oxides? (see [subtopic S3.1](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43116/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43116/>))

What about non-metal oxides? Are they acidic or basic? Think: Why does the release of oxides of nitrogen and sulfur into the atmosphere cause acid rain? (see [subtopic S3.1](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43116/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43116/>))

What are the conjugate acids of the polyatomic anions listed in [section S2.1.2](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ionic-bonding-id-43557/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ionic-bonding-id-43557/>)? Think that conjugate acids have only one more proton, not two! (see [subtopic S2.1](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43108/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43108/>))

Consolidate your understanding of conjugate acids and bases using the drag and drop activity in **Interactive 2**.

Drag the words into the correct boxes.

1. The conjugate base of OH^- is .
2. The conjugate base of H_2SO_3 is .
3. The conjugate acid of CO_3^{2-} is .
4. The conjugate acid of HPO_4^{2-} is .

✓ Check

Interactive 2. Understanding Conjugate Acids and Its Bases.

Amphiprotic substances

The reaction of ammonia with water showed that water behaved as an acid by donating a proton, while the reaction of hydrogen chloride gas with water showed that water behaved as a base by accepting a proton.

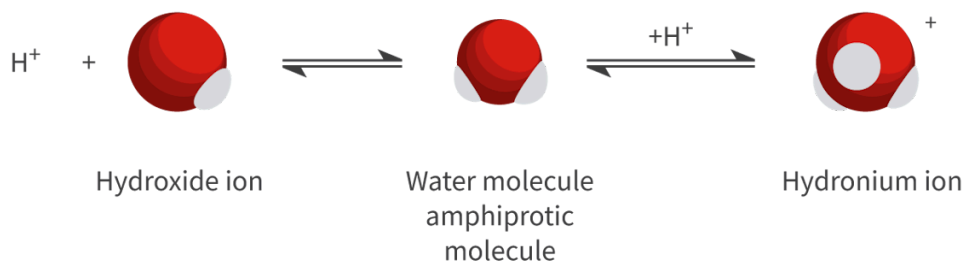


Figure 4. Water can both donate a proton and accept a proton.

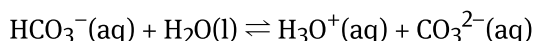
More information for figure 4

Species that are able to act as either a Brønsted–Lowry acid or a Brønsted–Lowry base are known as being amphiprotic. Water is such a substance.

Study skills

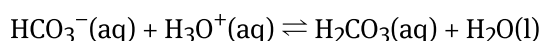
The term amphiprotic refers to acids and bases that are defined by the Brønsted–Lowry theory only. Another term, amphoteric, also refers to substances that can act as either acids or bases and applies to all acid–base theories.

The hydrogencarbonate ion (HCO_3^-) is an example of an amphiprotic species. This ion can lose a proton to form the carbonate ion (CO_3^{2-}), thereby acting as a Brønsted–Lowry acid:



In this reaction, HCO_3^- and CO_3^{2-} are a conjugate acid–base pair as they differ by a proton, as are H_2O and H_3O^+ .

But the hydrogencarbonate ion can also act as a Brønsted–Lowry base by gaining a proton to form H_2CO_3 :



In this reaction, HCO_3^- and H_2CO_3 are a conjugate acid-base pair with HCO_3^- being the base and H_2CO_3 being its conjugate acid.

Table 1 gives some examples of some amphoteric species.

Table 1. Some examples of amphoteric species.

Name of ion	Formula
Hydrogen carbonate ion	HCO_3^-
Hydrogen sulfate ion	HSO_4^-
Dihydrogen phosphate	H_2PO_4^-
Hydrogen phosphate	HPO_4^{2-}

Activity

- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 15 minutes
- **Activity type:** Pair activity

Instructions

Download the table provided and fill in the blanks with the corresponding formulas.

Table 

(https://d3vrb2m3yrmyfi.cloudfront.net/media/edusys_2/content_uploads/Chemistry3%20ACTIVITY%20worksheet.b21c96e1d0a1e71488ca.pdf)

Calculations involving pH and [H⁺]

Learning outcomes

By the end of this section you should be able to:

- (R3.1.4) Perform calculations involving the logarithmic relationship between pH and [H⁺].
- (R3.1.5) Recognise solutions as acidic, neutral and basic from the relative values of [H⁺] and [OH[−]].

You have probably heard the term pH before, perhaps in a previous science course or while reading some article. You might have heard statements that compare the acidity of acid rain to that of normal rain or tap water, using pH values. But what is pH, what does it measure and how is it calculated?

The pH scale

The pH scale is used to determine how acidic or basic an aqueous solution is. The pH of a substance is a measure of the concentration of hydrogen ions, [H⁺], in solution.

Table 1. Equation to calculate pH.

Equation	Symbols	Units
$\text{pH} = -\log[\text{H}^+(\text{aq})]$	pH = a measure of how acidic or basic a species is	No units
	$[\text{H}^+(\text{aq})]$ = concentration of aqueous hydrogen ions	Moles per cubic decim (mol dm ^{−3})

This equation can also be rearranged and expressed as:

$$[\text{H}^+] = 10^{-\text{pH}}$$

Study skills

The equation for pH can be found in [section 1](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/some-relevant-equations-id-45103/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/some-relevant-equations-id-45103/>) of the IBDP chemistry data booklet.

Recall that in [section S1.4.5a](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/molar-concentration-of-a-solution-id-44263/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/molar-concentration-of-a-solution-id-44263/>) we saw that square brackets are used to show the concentration of a substance. In this topic $[H^+]$ refers to the concentration of hydrogen ions in solution and $[OH^-]$ refers to the concentration of hydroxide ions in solution.

The pH scale is an inverse scale; the higher the concentration of H^+ ions in solution, $[H^+]$, the lower the pH value. Conversely, the lower the $[H^+]$, the higher the pH value. In terms of hydroxide ions, a higher concentration of OH^- ions, $[OH^-]$, corresponds to a higher pH value and a lower $[OH^-]$ corresponds to a lower pH value (**Figure 1**). Solutions commonly found in the school laboratory have a range of pH values from 0 (acidic) to 14 (alkaline). It should be noted that pH values below 0 and above 14 are possible.

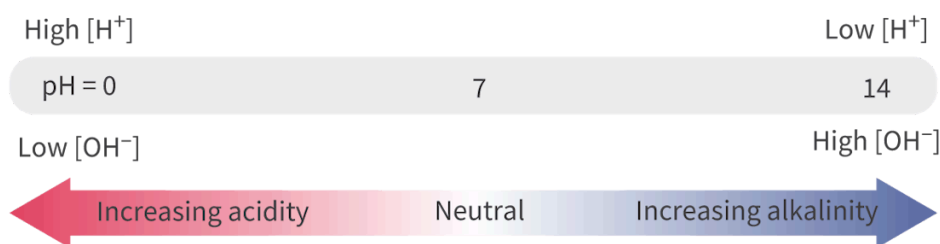


Figure 1. The pH scale is an inverse scale; the higher the $[H^+]$, the lower the pH value.

 More information for figure 1

The image is a diagram of the pH scale, illustrating the relationship between hydrogen ions (H^+) and hydroxide ions (OH^-) in a solution. The scale is depicted as a horizontal bar ranging from pH 0 to 14. On the left end of the scale, at pH 0, it indicates a high concentration of hydrogen ions and low concentration of hydroxide ions, marked as 'High $[H^+]$ Low $[OH^-]$ '. On the right end, at pH 14, it shows a low concentration of hydrogen ions and high concentration of hydroxide ions, labeled as 'Low $[H^+]$ High $[OH^-]$ '. The middle of the scale, pH 7, represents a neutral point. Below the scale, an arrow extends from left to right, transitioning in color from red on the left (indicating increasing acidity) to blue on the right (indicating increasing alkalinity), with the word 'Neutral' in the center below pH 7.

[Generated by AI]

The relationship between $[H^+]$ and $[OH^-]$ in aqueous solutions is shown in **Figure 2**. Note that the diagram refers to the indicator litmus, which we will discuss in more detail in [section R3.1.14-15](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/acid-base-indicators-hl-id-46062/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/acid-base-indicators-hl-id-46062/>).

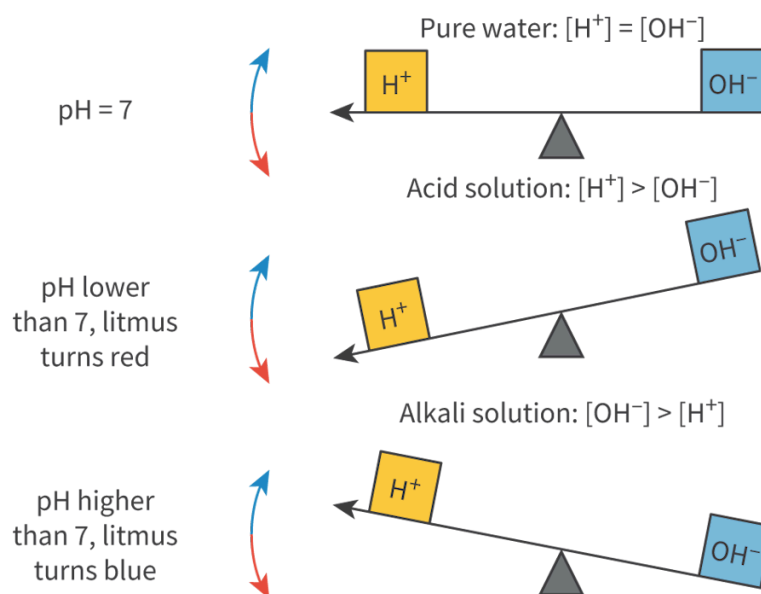


Figure 2. The balance between $[H^+]$ and $[OH^-]$ in aqueous solutions.

 More information for figure 2

The diagram illustrates the balance between hydrogen ions (H^+) and hydroxide ions (OH^-) using a see-saw analogy. It consists of three scenarios:

1. At the top, the see-saw is level, representing neutral conditions where $[H^+]$ equals $[OH^-]$, typical of pure water. The text notes that the pH is 7.
2. The middle section shows the see-saw tilted to the left, indicating acidic conditions where $[H^+]$ is greater than $[OH^-]$. This section notes that pH is lower than 7, and litmus turns red.
3. At the bottom, the see-saw tilts to the right, representing alkaline conditions where $[OH^-]$ is greater than $[H^+]$. It is stated that pH is higher than 7, and litmus turns blue.

Each scenario is accompanied by rectangular labels representing H^+ and OH^- . Arrows beside the see-saws depict directionality correlated with the see-saw balance, indicating the shifts in acidity or alkalinity.

[Generated by AI]

If the concentration of hydrogen ions is equal to the concentration of hydroxide ions, the solution is neutral. Pure water is an example of a neutral substance because $[H^+] = [OH^-]$. If the $[H^+]$ is greater than the $[OH^-]$, the solution is acidic and if the $[OH^-]$ is greater than the $[H^+]$, the solution is basic, or alkaline. The relationship between $[H^+]$ and $[OH^-]$ is summarised in the **Table 1**.

Table 1. Acidic, basic or neutral?

	Acidic, basic or neutral	pH at 298 K
$[H^+] = [OH^-]$	neutral	7
$[H^+] > [OH^-]$	acidic	<7
$[OH^-] > [H^+]$	basic	>7

From the mathematical definition of pH, you should understand that this scale is logarithmic to the base 10. This means that a change of one pH unit corresponds to ten times the change in hydrogen ion concentration. For example, an aqueous solution with a pH = 3 is 10 times more acidic than an aqueous solution with a pH = 4; and 100 times (or 10×10) more acidic than an aqueous solution with a pH = 5.

Practical skills

Tool 3: Mathematics — Carry out calculations involving logarithmic functions, Use and interpret scientific notation

Remember:

- A logarithmic value such as pH should have the same number of decimal places as the least number of significant figures in the values used to calculate it.
- When in doubt with significant figures, it is always useful to convert the numbers to scientific notation.

Table 2 shows some values that emphasise the relationship between the concentration of hydrogen ions and pH. Not all the values are shown, but you should be able to work them out from the pattern.

Table 2. The relationship between $[H^+]$ concentration and pH.

pH	$[H^+(aq)] / \text{mol dm}^{-3}$
0	1×10^0 or 1
1	1×10^{-1} or 0.1
2	1×10^{-2} or 0.01

pH	[H ⁺ (aq)] / mol dm ⁻³
6	1×10^{-6} or 0.000001
7	1×10^{-7} or 0.0000001
8	1×10^{-8} or 0.00000001
13	1×10^{-13} or 0.00000000000001
14	1×10^{-14} or 0.000000000000001

Table 2 demonstrates that if the pH of a solution is given the value x , then this corresponds to a hydrogen ion concentration of 10^{-x} mol dm⁻³.

Study skills

A change in one pH unit represents a tenfold change in [H⁺]. An increase of two pH units, for example, represents a 100 times change in [H⁺].

Hence, a solution with a pH = 1 has 10 times more hydrogen ions than a solution with pH = 2, and 100 times more than a pH = 3.

Every time that we want to state the strength of an acid we need to express the [H⁺] using scientific notation or write out a string of zeros. If, however, we use the base 10 logarithm (log₁₀) of very large or very small numbers, we can avoid the use of scientific notation and it also allows us to compare the concentration of hydrogen ions in solution more easily.

Nature of Science

Aspect: Theories, Models

Why use a pH scale — Occam's razor.

The pH scale is an attempt to scale the relative acidity over a wide range of H⁺ concentrations into a very simple number. It is an example of applying Occam's razor.

Occam's razor is the philosophical principle that the simplest scientific explanation which fits all known facts is the best one.

Sørensen's idea behind the pH scale was just to find a convenient way of expressing a very wide range of numbers in a simpler-to-write form. He used a logarithmic scale — it is a convenience linked to the reality of the concentration of hydrogen and hydroxide ions in pure water at 25°C.

Scientists often use a logarithmic scale when there is a wide range of values to be expressed and considered. We have seen this when plotting the successive first ionisation energies of atoms and more generally on the Richter scale associated with measuring the strength of earthquakes.

Using such a scale is useful when the significance of a change in value lies in its relation to the value itself, rather than any absolute size. pH is defined as the negative of the logarithm of the concentration of hydrogen ions. The use of a logarithmic scale eases the written presentation of the figures, and also their graphical display. Otherwise, for instance, we would be writing the [H⁺] in neutral water as 0.0000001 mol dm⁻³ (or 10⁻⁷ mol dm⁻³) — both of which are less easy on the eye than pH 7.0. The use of the term pH aids the discussion of values and also their graphical presentation.

pH calculations

From the definition of pH, it is possible to calculate the pH value of a solution from its [H⁺]. For strong acids or bases, the [H⁺] or [OH⁻] in solution is equal to the initial concentration of the strong acid or base. This is because strong acids and bases completely dissociate in solution. Rearranging the equation for pH allows us to calculate the concentration of H⁺ ions in a solution from its pH value:

$$\text{pH} = -\log[\text{H}^+(\text{aq})]$$

$$[\text{H}^+(\text{aq})] = 10^{-\text{pH}}$$

Worked example 1

Calculate the pH of 0.020 mol dm⁻³ nitric acid, HNO₃(aq).

Solution steps	Calculations
Step 1: Write out the values given in the question.	$[\text{H}^+] = 0.020 \text{ mol dm}^{-3}$ Note that [H ⁺] is equal to the initial conc because it is a strong acid that is comple
Step 2: Write out the equation.	$\text{pH} = -\log_{10}[\text{H}^+]$
Step 3: Substitute the values given.	$= 1.70$

Solution steps	Calculations
Step 4: State the answer with appropriate units and the number of significant figures used in rounding.	Since the digits to the left of the decimal exponential function, they do not represent significant figures. Therefore, only the digits to the right of the decimal point in the measurement are significant.

Worked example 2

A solution has [H⁺] of $1.0 \times 10^{-8} \text{ mol dm}^{-3}$. Calculate its pH value.

Solution steps	Calculations
Step 1: Write out the values given in the question.	[H ⁺] = $1.0 \times 10^{-8} \text{ mol dm}^{-3}$
Step 2: Write out the equation.	pH = $-\log_{10} [\text{H}^+]$
Step 3: Substitute the values given.	= $-\log_{10} (1.0 \times 10^{-8})$
Step 4: State the answer with appropriate units and the number of significant figures used in rounding.	= 8.00 Recall units.

Worked example 3

Calculate the [H⁺] of a solution with a pH value of 5.30.

Solution steps	Calculations
Step 1: Write out the values given in the question.	pH = 5.30
Step 2: Write out the equation.	$[\text{H}^+] = 10^{-\text{pH}}$
Step 3: Substitute the values given.	= $10^{-5.30}$

Solution steps	(
Step 4: State the answer with appropriate units and the number of significant figures used in rounding.	=

Acid-base indicators

In sections R3.1.8 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ph-curves-for-strong-acid-base-neutralisation-reactions-id-46057/>) and R3.1.14–15 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/acid-base-indicators-hl-id-46062/>), you will learn that an indicator changes colour depending on the pH of the solution. Two common indicators, litmus paper and universal indicator, are discussed below.

Litmus paper

Litmus paper is a common indicator used in the school laboratory. Like many indicators, it is made from a natural source, lichen. Litmus paper comes in two colours: red and blue. In an acidic solution, blue litmus changes to red and in an alkaline solution, red litmus changes to blue.

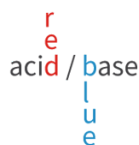


Figure 3. The colours of litmus paper in acidic and basic solutions.

 More information for figure 3

The diagram illustrates the color changes of litmus paper based on the pH of the solution. The text "red" is in red color oriented vertically, representing the change of blue litmus paper to red in acidic conditions. Adjacent to it is the word "acid" in gray, centrally positioned. To the right, in blue, vertically oriented, is the text "blue," indicating the change of red litmus paper to blue in basic conditions. Below "blue" is the word "base" in gray. The diagram visually explains the basic and acidic changes in litmus paper.

[Generated by AI]

Other indicators give different colours in acidic and alkaline solutions. Some common examples are given in **Table 3**.

Table 3. Some common examples of acid-base indicators and their colours.

Indicator	Colour in acid	Colour in alkali
Litmus	red	blue
Methyl orange	red	yellow
Phenolphthalein	colourless	pink/purple

Although litmus is a very commonly used indicator, its usefulness is limited as it is only a qualitative indicator – it can detect if something is acidic or basic but it cannot determine the pH value. This is also true of the other indicators that just have a single colour change.

Universal indicator

Universal indicator is a quantitative indicator as it is actually a mixture of indicators that produces different colours depending on the pH of the solution. The mixture is designed so that the colours produced as the pH increases usually correspond to a 'rainbow' sequence as shown in **Figure 4**. The pH of aqueous solutions can be measured by using universal indicator as either a solution or in paper form.

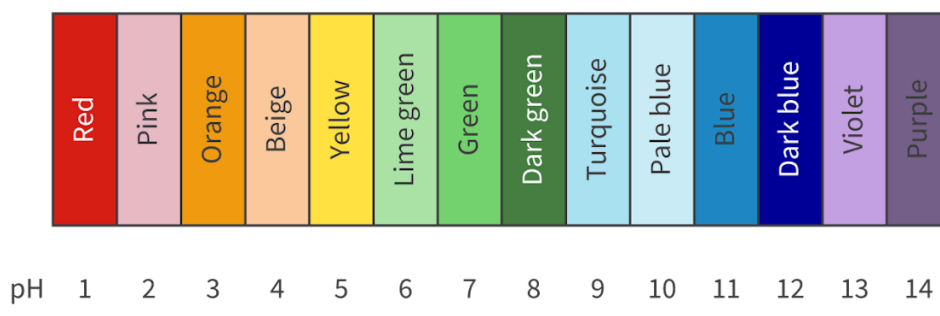


Figure 4. The pH scale and the universal indicator colours.

 More information for figure 4

The image displays a pH scale ranging from 1 to 14. Beneath each pH value, there is a corresponding color which represents the universal indicator color for that pH level. Starting from pH 1, the sequence of colors includes red, pink, orange, beige, yellow, lime green, green, dark green, turquoise, pale blue, blue, dark blue, violet, and ends with purple at pH 14. The colors transition smoothly, creating a rainbow-like gradient that helps to identify pH levels based on color changes.

[Generated by AI]

Indicators can be used to measure the pH of a solution, by making use of the fact that their colour changes with pH. Visual comparison of the colour of a test solution with a standard colour chart allows us to measure the pH to the nearest whole number.



Figure 5. Reading the pH of a solution by matching to the colour chart.

Credit: Tetiana Garkusha, Getty Images

 More information for figure 5

The image shows a hand holding two small vials containing colored solutions. The vials are placed over a pH color chart consisting of multiple colored squares. Each square on the chart is labeled with a pH value, ranging from acidic (red, orange) through neutral (yellow, green) to basic (dark green, blue). The vial on the left contains a yellow solution, while the vial on the right contains an orange solution. The hand is individually matching each vial to the closest color on the chart to determine the pH value of the solutions.

[Generated by AI]

Worked example 4

A substance has a pink colour with phenolphthalein and blue colour with litmus. Using **Table 3**, identify if the substance is an acid or a base.

The substance is a base since the colour of litmus with a base is blue and the colour of phenolphthalein is pink.

Practical skills

Tool 1: Experimental techniques — Measuring variables, pH of a solution

Tool 2: Technology — Applying technology to collect data

Can you think of a more accurate way to determine pH other than the use of universal indicator?

A pH probe and meter produces a more accurate method of measuring pH, provided they are correctly calibrated (**Figure 6**). The electrode of the pH meter is placed in the solution to be tested and a voltage is generated that is converted into a pH meter reading displayed on the screen. The pH meter is calibrated using buffers of known pH, usually of pH 4.0, 7.0 and 10.0.

For practical work you might use the pH probe or meter.

You will measure the pH of prepared solutions to determine their acidity, or determine pH at the equivalence point of a titration. The values of pH will indicate to you if a solution is acidic or basic and you would be able to derive information regarding the substances that are in excess or if a reaction is complete or not.



Figure 6. A pH meter reads the pH of a solution electronically.

Credit: photongpix, Getty Images

Aspect: Experiments

The relative strengths of acids and bases can now be measured using improved instrumentation.

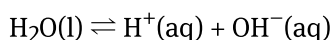
Alongside the development of theories of acid base behaviour, the means of measuring acidity were also being enhanced. The concept of pH was an important step in that it allowed comparisons to be made between solutions of different acids at the same concentration. While no absolute comparison of the strengths of acids could be made, this enabled the relative strengths of different acids to be compared.

Technical and theoretical considerations went hand-in-hand. In 1889, Nernst, a German chemist, provided a theoretical foundation for using electrode potential to measure the ionic concentration of a solution. Then Sørensen, in 1900, developed the pH scale. Not only that, but he also devised an early form of pH meter using a hydrogen electrode combined with a calomel reference electrode.

Such a device added to the methods for estimating pH that used various indicators, including litmus. Then, in 1934, pH measurement was revolutionised by the development of the modern pH meter by Arnold Beckman. This instrument was originally developed for citrus fruit growers in California to monitor fruit acidity during the production of pectin and citric acid.

The ionic product of water

Pure water is a very poor conductor of electricity, which means that it must have a low concentration of mobile ions responsible for the electrical conductivity of solutions. Water molecules do dissociate however, but only to a very small extent. This is known as the auto-ionisation of water. This reaction can be represented by the following equation:



The equilibrium constant expression (K_c) for the above reaction is:

$$\begin{aligned} K_c &= K_w \\ &= [\text{H}^+][\text{OH}^-] \end{aligned}$$

which is known as the ionic product of water. The ionic product of water is the product of the $[\text{H}^+]$ and $[\text{OH}^-]$ in water at a particular temperature. At 298 K (25°C), pure water has $[\text{H}^+]$ and $[\text{OH}^-]$ of $1.00 \times 10^{-7} \text{ mol dm}^{-3}$. At 298 K, the value of K_w for pure water can be calculated as follows:

$$\begin{aligned} K_w &= [\text{H}^+][\text{OH}^-] \\ &= 1.00 \times 10^{-7} \times 1.00 \times 10^{-7} \\ &= 1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6} \end{aligned}$$

Study skills

The equation for K_w and its value at 298 K are given in the DP chemistry data booklet in [section 1 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/some-relevant-equations-id-45103/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/some-relevant-equations-id-45103/).

Like any equilibrium expression seen in [subtopic R2.3](#)

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43482/>), K_w is temperature-dependent, therefore, at a certain temperature the product of [H⁺] and [OH⁻] is a constant. This means that as [H⁺] of a solution increases, [OH⁻] decreases (and vice versa). A neutral solution is any which has [H⁺] equal to [OH⁻]. Recall that acidic solutions have [H⁺] > [OH⁻] and basic solutions have [OH⁻] > [H⁺].

Making connections

Can you make a connection to Le Châtelier's principle and think why does the ionisation of water increase as temperature increases? (see [subtopic R2.3](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43482/>)).

Calculating [H⁺] and [OH⁻]

Using the value of the ionic product of water, the [H⁺] and [OH⁻] of a solution can be calculated.

Worked example 5

An aqueous solution has a pH of 3.20 at 298 K.

Determine the [H⁺] and [OH⁻] of the solution.

Solution steps	Calculations
Step 1: Write out the values given in the question.	pH = 3.20
Step 2: Write out the equation.	[H ⁺] = 10 ^{-pH}
Step 3: Substitute the values given.	$= 10^{-3.20}$ $= 6.31 \times 10^{-4} \text{ mol dm}^{-3}$
Using the [H ⁺] in solution and the K_w , we can determine the [OH ⁻] in	
Step 4: Write out the required values.	$[H^+] = 6.31 \times 10^{-4}$ $K_w = 1.00 \times 10^{-14}$

Solution steps	Calculations
Step 5: Write out the equation.	$K_w = [\text{H}^+][\text{OH}^-]$
Step 6: Rearrange the equation.	$[\text{OH}^-] = \frac{K_w}{[\text{H}^+]}$
Step 7: Substitute the values given.	$[\text{OH}^-] = \frac{1.00 \times 10^{-14}}{6.31 \times 10^{-2}}$
Step 8: State the answer with appropriate units and the number of significant figures used in rounding.	$= 1.58 \times 10^{-11} \text{ mol dm}^{-3}$

Worked example 6

Determine the pH of a $0.084 \text{ mol dm}^{-3}$ solution of lithium hydroxide, LiOH at 298 K.

First calculate the value of $[\text{H}^+]$.

Lithium hydroxide is a strong base, therefore, it is completely ionised.

Solution steps	Calculations
Step 1: Write out the values given in the question.	$[\text{OH}^-] = 0.084 \text{ mol dm}^{-3}$ $K_w = 1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$
Step 2: Write out the equation.	$K_w = [\text{H}^+][\text{OH}^-]$
Step 3: Rearrange the equation.	$[\text{H}^+] = \frac{K_w}{[\text{OH}^-]}$
Step 4: Substitute the values given.	$[\text{H}^+] = \frac{1.00 \times 10^{-14}}{0.084}$ $= 1.19 \times 10^{-13} \text{ mol dm}^{-3}$

Now calculate the pH of the solution.

Solution steps	Calculations
Step 5: Write out the required values.	$[H^+] = 1.19 \times 10^{-13} \text{ mol dm}^{-3}$
Step 6: Write out the equation.	$\text{pH} = -\log[H^+]$
Step 7: Substitute the values given.	$\text{pH} = -\log(1.19 \times 10^{-13})$ $= 12.92$
Step 8: State the answer with appropriate units and the number of significant figures used in rounding.	$= 13 \text{ (2 s.f.)}$

Use the flashcards in **Interactive 1** to specify if the solution is acidic or basic based on the information given.

Specify if the solution is acidic or basic based on the given information.

pH is 3

 Turn

Card 1 of 4 >

Interactive 1. Review Activity for Acidic Basic Determination Based on given Values.

Try the activity in the next section to check your understanding of pH.

Strong and weak acids and bases

Learning outcomes

By the end of this section you should be able to recognise that acid–base equilibria lie in the direction of the weaker conjugate.

If you measure the pH of 0.1 mol dm^{-3} solutions of lemon acid and battery acid you would obtain different values. The pH of lemon juice would be around 2–3 while that of battery acid would be around 0–1. Why? What is the difference between the two acids?

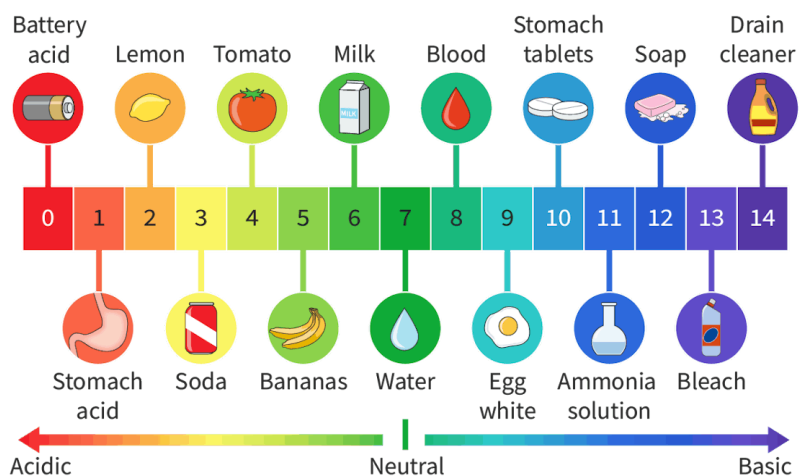


Figure 1. The pH of battery acid and lemon juice are different.

More information for figure 1

The image shows a pH scale ranging from 0 to 14, marking the acidic, neutral, and basic regions. At pH 0, battery acid is depicted with an image of a battery. Lemon juice, with a lemon icon, corresponds to pH 2. Other substances shown include tomato at pH 4, milk at pH 6, and blood at pH 7, the neutral point. The diagram continues with substances like stomach tablets at pH 9, soap at pH 12, and drain cleaner at pH 14. An arrow at the bottom indicates the progression from acidic to basic.

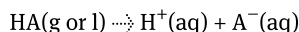
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Acids and bases can be classified as being strong or weak. The strength of an acid refers to its degree of dissociation (or ionisation) in aqueous solution. Strong acids and bases completely dissociate (or ionise) in solution whereas weak acids and bases only partially dissociate (or ionise) in solution.

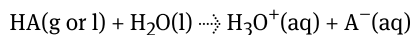
Strong and weak acids

We saw in section R3.1.4-5a ([https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculations-involving-ph-and-\[h+\]-id-44702/](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculations-involving-ph-and-[h+]-id-44702/)) that when a strong acid dissolves in water, it completely dissociates into ions; that is, all the molecules of acid react with the water to produce hydronium ions (H_3O^+), or

hydrogen ions (H^+). The general equation for the dissociation of a strong acid, HA, is given below. Note that a one-way arrow ($\rightarrow$) is used to show the complete dissociation (or ionisation) of the acid.



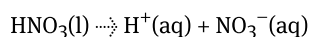
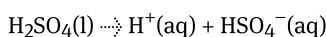
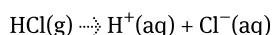
This reaction can also be written with the hydronium ion (H_3O^+) included:



Study skills

For a strong acid, the concentration of H^+ ions is equal to the initial concentration of the strong acid.

The equations for the dissociation of the three common strong acids, hydrochloric, sulfuric and nitric acids, are shown below:



Note that sulfuric acid, H_2SO_4 , is diprotic (it contains two acidic hydrogens). However, only the first dissociation can be considered strong.

Study skills

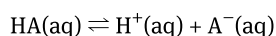
HCl, HBr, HI, HNO_3 and H_2SO_4 are strong acids.

Organic acids, such as ethanoic acid, are usually weak acids. When ethanoic acid is added to water, only a relatively small percentage (1% or less) of its molecules dissociate to form H_3O^+ ions. An equilibrium is established in which the majority of the acid molecules do not dissociate into ions. In other words, the equilibrium position lies very much to the left and the value of the equilibrium constant, K_c , for the dissociation is small.

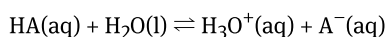
Study skills

For a weak acid (or weak base) the equilibrium position lies to the left and the value of the equilibrium constant, K_c , for the dissociation is small.

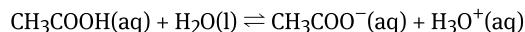
The general equation for the dissociation of a weak acid HA is shown below. Note the use of the equilibrium sign ($\rightleftharpoons$) in the equation.



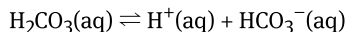
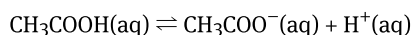
This reaction can also be written involving the formation of the hydronium ion (H_3O^+):



Common examples of weak acids are ethanoic acid (CH_3COOH) and carbonic acid (H_2CO_3), which dissociate according to the following equations:

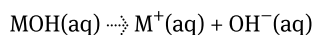


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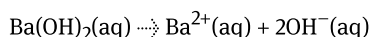
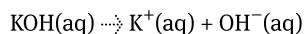
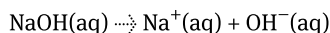


Strong and weak bases

A strong base (or alkali) is one which completely dissociates (or ionises) into ions in aqueous solution. Strong bases include the metal hydroxides of group 1 (LiOH and NaOH) and barium hydroxide ($\text{Ba}(\text{OH})_2$). The general equation for the dissociation of a strong base, MOH , is shown below.



The three common strong bases are sodium hydroxide, potassium hydroxide and barium hydroxide, which dissociate according to the following equations:



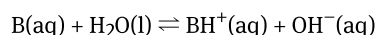
Like strong acids, the equilibrium position for the dissociation of a strong base lies very much to the right. For this reason, a 1 mol dm^{-3} solution of NaOH will contain 1 mol dm^{-3} of OH^- ions. Note that the dissociation of one mole of barium hydroxide, $\text{Ba}(\text{OH})_2$, produces two moles of OH^- ions.

Study skills

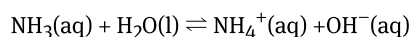
Group 1 hydroxides are strong bases.

For a strong base, the concentration of OH^- ions, $[\text{OH}^-]$, is equal to the initial concentration of the strong base.

All other bases, except for the hydroxides of groups 1 and 2, are weak. Weak bases are made up of molecules that partially dissociate in water to release hydroxide ions. The general equation for the dissociation of a weak base, B , is shown below.



At equilibrium, the majority of the weak base molecules have not dissociated into ions. In other words, the equilibrium position lies very much to the left and the value of the equilibrium constant, K_c , for the dissociation is small. Examples of weak bases include aqueous ammonia (NH_3) and amines such as ethanamine ($\text{C}_2\text{H}_5\text{NH}_2$). The equations for the dissociation of these weak bases are shown below.



Study skills

When defining a strong or weak acid or base, use the terms complete and partial to describe their dissociation (or ionisation). Also, remember to use an equilibrium sign ($\rightleftharpoons$) for weak acids and bases.

For weak acids and bases, the equilibrium concentrations of H_3O^+ ions, or OH^- ions, are much lower than for strong acids and bases. These differences in concentration can be used to distinguish between solutions of weak and strong acids and bases.

The calculations involved in determining the pH of a strong acid were covered in section R3.1.4–5a ([https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculations-involving-ph-and-\[h+\]-id-44702/](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculations-involving-ph-and-[h+]-id-44702/)). Recall that when calculating the pH of a strong acid or base, the $[\text{H}^+]$ or $[\text{OH}^-]$ are equal to the initial concentration of the strong acid or base.

Distinguishing between weak and strong acids and bases

Practical skills

Tool 1: Experimental techniques — Measuring variables

Inquiry 3: Collecting and processing data — Interpreting results

Now that you know the difference between weak and strong acids and bases, discuss with a partner what physical and chemical properties can be observed in the lab to distinguish between weak and strong acids or bases of the same concentration?

Study skills

When comparing the properties of weak and strong acids or bases, make sure that you are comparing solutions of equal concentration. Concentration and strength are two different variables that together will determine the pH of the solution. Concentration is the ratio of solute:solution, while strength is the degree of ionisation. For example, you can have two solutions — a high concentration of a weak acid and a low concentration of a strong acid — that can have the same pH.

Table 1. A summary of the methods described for strong and weak acids.

	$0.1 \text{ mol dm}^{-3} \text{ HCl}(\text{aq})$	$0.1 \text{ mol dm}^{-3} \text{ CH}_3\text{COOH}(\text{aq})$
$[\text{H}^+ (\text{aq})]$	0.1 mol dm^{-3}	$0.0013 \text{ mol dm}^{-3}$
pH	1.00	2.87
Electrical conductivity	high	low

	0.1 mol dm ⁻³ HCl(aq)	0.1 mol dm ⁻³ CH ₃ COOH(aq)
Relative rate of reaction with magnesium	fast	slow
Relative rate of reaction with calcium carbonate	fast	slow

How can you differentiate between strong and weak bases?

Strong bases have higher pH values and electrical conductivity than weak bases for solutions of the same concentration.

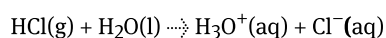
Table 2. Differentiating between strong and weak bases.

	0.1 mol dm ⁻³ NaOH(aq)	0.1 mol dm ⁻³ NH ₃ (aq)
[OH ⁻ (aq)]	0.1 mol dm ⁻³	0.0013 mol dm ⁻³
pH	13–14	10–11
Electrical conductivity	high	low

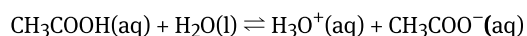
Strength of conjugate acids and bases

We will now look at the strength of the conjugate bases in the reactions of hydrochloric acid (a strong acid) and ethanoic acid (a weak acid) with water.

In the reaction with hydrochloric acid, the water molecule is a much stronger base than the chloride ion. In other words, the water molecule has a much greater tendency to accept a proton than the chloride ion. Therefore, the position of the equilibrium lies to the right and virtually all of the hydrogen chloride molecules will dissociate.



In the reaction of ethanoic acid with water, the ethanoate ion is a much stronger base than the water molecule. The ethanoate ion has a much greater tendency to accept a proton than the water molecule. Therefore, the position of equilibrium lies to the left and relatively few ethanoic acid molecules will dissociate.



In general, strong acids have relatively weak conjugate bases and weak acids have relatively strong conjugate bases.

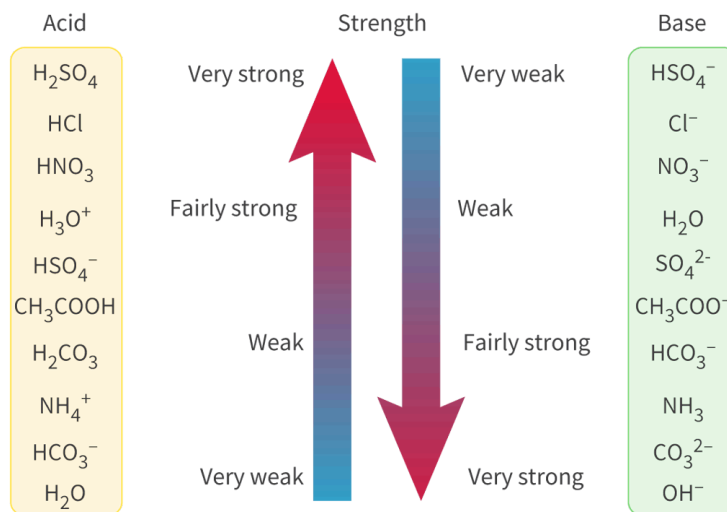


Figure 2. Some common acids and conjugate bases in order of their strengths.

More information for figure 2

The diagram illustrates a comparison of acids and their conjugate bases with labels indicating their relative strengths. On the left side, a column labeled 'Acid' lists various acids: H₂SO₄, HCl, HNO₃, H₃O⁺, HSO₄⁻, CH₃COOH, H₂CO₃, NH₄⁺, HCO₃⁻, and H₂O. On the right side, a column labeled 'Base' lists the corresponding conjugate bases: HSO₄⁻, Cl⁻, NO₃⁻, H₂O, SO₄²⁻, CH₃COO⁻, HCO₃⁻, NH₃, CO₃²⁻, and OH⁻.

Between these columns are two large vertical arrows labeled 'Strength,' depicting the relative acid and base strengths. The left arrow indicates acids range from 'Very strong' at the top to 'Very weak' at the bottom. The right arrow indicates conjugate base strengths listed from 'Very weak' at the top to 'Very strong' at the bottom. The arrow direction shows the inverse relationship between acid and conjugate base strength.

[Generated by AI]

A strong acid will ionise more than a weak acid and the position of equilibrium lies to the right; the conjugate base is relatively weak. Conversely, for a weak acid, the conjugate base is relatively strong, and the equilibrium lies to the left. A similar situation occurs for bases. A strong base has a weak conjugate acid and a weak base has a strong conjugate acid.

Making connections

Knowing that strong acids completely dissociate, while weak acids partially dissociate, how would you expect the equilibrium constants of strong and weak acids to compare? (see [subtopic R2.3](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43482/>)).

Think about the bonds involved, and based on your knowledge of strong and weak acids, why does the acid strength of the hydrogen halides increase down group 17? (see [subtopic R1.1](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43468/>)).

Creativity, activity, service

Strand: Service

Learning outcomes:

- Demonstrate engagement with issues of global significance
- Demonstrate how to initiate and plan a CAS experience

Have you ever considered the effects that acids can have on the environment?

One such example is acid rain. The combustion of coal in power stations can produce acidic gases such as sulfur dioxide (SO_2). Once in the atmosphere, the SO_2 dissolves in water to produce acid rain. Acid rain can have harmful effects

 ([https://www.eia.gov/energyexplained/coal/coal-and-the-environment.php#:~:text=Several%20principal%20emissions%20result%20from, respiratory%20illnesses%20and?on the natural environment and can also damage statues, particularly those made of limestone \(calcium carbonate\).](https://www.eia.gov/energyexplained/coal/coal-and-the-environment.php#:~:text=Several%20principal%20emissions%20result%20from, respiratory%20illnesses%20and?on%20the%20natural%20environment%20and%20can%20also%20damage%20statues,particularly%20those%20made%20of%20limestone%20(calcium%20carbonate).))

A possible CAS experience could be to research the use of coal in your country or region and raise awareness about the effects of its use as an energy source.

Try the activity in the next section to check your understanding of acids and bases.

Reactions of acids

Learning outcomes

By the end of this section you should be able to formulate equations for the reactions between acids and metal oxides, metal hydroxides, hydrogencarbonates, carbonates, ammonia and amines.

Acids are often thought of in a negative way like in the case of acid rain, but there are many acids that are helpful to us. Can you think of any?

Two examples are stomach acid is needed to break down the food we eat to allow for chemical digestion to take place (**Figure 1**) and acidic cleaning products that are used to kill bacteria.

Some of these reactions are explored in this section.

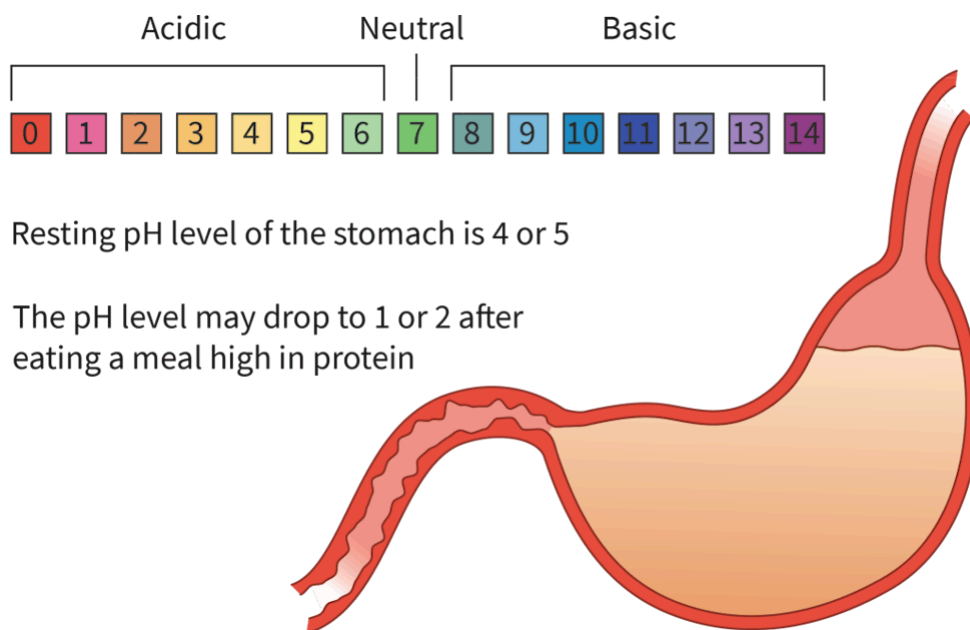


Figure 1. pH levels of the stomach before and after a meal.

 More information for figure 1

The image is a diagram featuring an illustration of a stomach and a pH scale. The pH scale at the top is segmented into acidic, neutral, and basic regions, ranging from 0 to 14. The numbers 0 through 6, colored in gradients of red to yellow, represent acidic pH

levels, with 0 being highly acidic. Number 7 is marked as neutral and is colored green. Numbers 8 through 14, colored in shades of blue to purple, represent basic pH levels.

Below the pH scale, two statements explain the pH levels concerning the stomach. The first statement indicates that the resting pH level of the stomach is 4 or 5, which falls into the acidic range of the scale. The second statement explains that the pH level may drop to 1 or 2 after consuming a high-protein meal, indicating increased acidity. The illustration of the stomach is shown beneath these statements, highlighting its shape and location in the digestive system.

[Generated by AI]

What happens to the pH of an acid (low pH) as it reacts with a base (high pH)?

The pH will move towards the middle, closer to neutral. Consequently, these reactions are known as neutralisation reactions.

In this section we will look into the reactions of acids with these bases:

- metal oxides and metal hydroxides
- hydrogencarbonates and carbonates
- ammonia and amines.

Study skills

Recall from section R3.1.1 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/br%C3%B8nsted-lowry-acids-and-bases-id-43485/>), that the distinction between the terms 'base' and 'alkali' should be understood:

Bases that are insoluble in water are simply called bases. Examples of bases include the metal oxides calcium oxide and copper oxide.

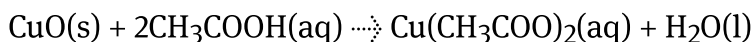
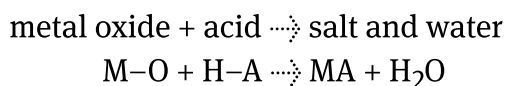
Bases that are soluble in water are known as alkalis. Examples of alkalis include the group 1 metal hydroxides, sodium hydroxide and lithium hydroxide.

Reaction with metal oxides

Acids undergo neutralisation reactions with metal oxides to produce a salt and water. This is a displacement reaction that will form a predictable set of products – the ionic salt and water. The reaction involves:

- The metal cation (M^+) in the oxide bonding with the anion (A^-) in the acid to form the salt (MA).
- The hydrogen (H^+) in the acid bonding with the oxygen (O) in the base to form water (H_2O).

The general word equation is given below, together with some specific examples:

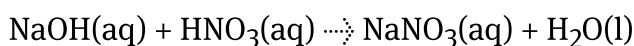
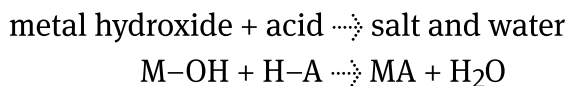


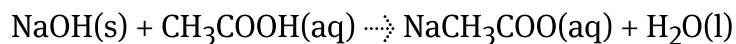
Reaction with metal hydroxides

Acids undergo a similar reaction with metal hydroxides to produce a salt and water. The reaction involves:

- The metal cation (M^+) in the hydroxide bonding with the anion (A^-) in the acid to form the salt (MA).
- The hydrogen ion (H^+) in the acid bonding with the hydroxide (OH^-) in the base to form water (H_2O).

The general word equation is given below, together with some specific examples:





The salt produced in the reaction depends on the parent acid and base that react (**Figure 2**). For example, the reaction of sodium hydroxide and nitric acid produces sodium nitrate.

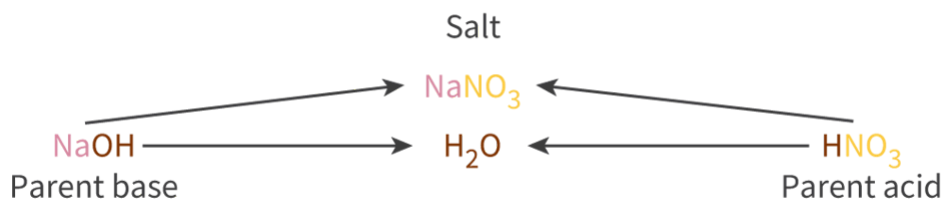


Figure 2. The make-up of a salt is defined by the acid and base it is made from.

More information for figure 2

The diagram illustrates how a salt is formed from the reaction between a parent acid and a parent base. On the left side, the parent base is represented as NaOH , and on the right side, the parent acid is HNO_3 . Arrows point from each reactant toward a central product labeled as NaNO_3 , which is the salt. Below the salt, another arrow points downwards to H_2O , indicating the formation of water. This schematic represents the chemical equation where sodium hydroxide (NaOH) reacts with nitric acid (HNO_3) to produce sodium nitrate (NaNO_3) and water (H_2O).

[Generated by AI]

Worked example 1

Determine the products of these reactions and complete the balanced equations.

1. $\text{Mg(OH)}_2(\text{aq}) + \text{H}_2\text{SO}_4(\text{aq})$
2. $\text{CaO(s)} + \text{HNO}_3(\text{aq})$

First identify the substances involved, then write the formulas of the products, then balance the equations.





Worked example 2

Identify the acid and base that produce the following salts:

1. Li_2SO_4
2. $\text{Mg(NO}_3)_2$
3. $\text{CH}_3\text{CH}_2\text{COOK}$
4. CaCl_2

The metal cation is from a base (combine with OH^-) and the anion is from the acid (combine with H^+).

1. LiOH and H_2SO_4
2. Mg(OH)_2 and HNO_3
3. KOH and $\text{CH}_3\text{CH}_2\text{COOH}$
4. Ca(OH)_2 and HCl

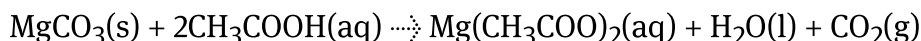
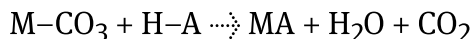
Reaction with metal carbonates

Acids react with metal carbonates to produce a salt, carbon dioxide gas and water. The reaction involves:

- The metal cation (M^+) in the carbonate bonding with the anion (A^-) in the acid to form the salt (MA).
- The hydrogen (H^+) in the acid together with the carbonate ion forming two products – carbon dioxide (CO_2) and water (H_2O).

The general word equations are given below, together with some specific examples.

metal carbonate + acid $\rightarrow$ salt + water + carbon dioxide



The presence of carbon dioxide gas can be tested for by bubbling the gas through an aqueous solution of calcium hydroxide, $\text{Ca}(\text{OH})_2(\text{aq})$, otherwise known as limewater. If carbon dioxide gas is present, the limewater turns a 'milky' colour as a solid precipitate of calcium carbonate is formed (**Video 1**).

Testing for CO₂ (Carbon dioxide) with Limewater



Video 1. Demonstration of the limewater test.

Can you correlate the reaction occurring in the limewater to an acid-base reaction?

Remember that carbon dioxide increases the acidity of water, and limewater is a basic solution.

Worked example 3

Complete and balance the equation for the reaction between solid sodium carbonate and hydrochloric acid:



Identify first the type of substances involved, then write their formulas and those of the products, then balance the equations.



International Mindedness

Coral reefs appear to be made of rock, but are actually colonies of animals called polyps. The polyps have a reciprocal relationship with the plant life in the ocean, providing carbon dioxide and waste products which are used for photosynthesis. In return, the plants provide oxygen and nutrients to the polyps, allowing them to make their calcium carbonate skeletons. Carbon dioxide in the air dissolves into the ocean, forming carbonic acid. The steady increase in carbon dioxide that comes from the burning of fossil fuels has caused an increase in the concentration of carbonic acid, which then reacts with the calcium carbonate skeletons in a neutralisation reaction. Globally, this has resulted in a 40–50% decrease in coral over the last 30 years. Coral reefs provide protection to the shoreline, a habitat for marine organisms and limestone for construction. All countries are connected by the air we breathe and the water we share, threatening the survival of coral reefs and the benefits we gain from them.

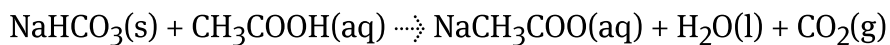
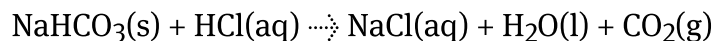
Reaction with metal hydrogencarbonates

Acids react with metal hydrogencarbonates in a very similar reaction. The reaction involves:

- The metal cation (M^+) in the carbonate bonding with the anion (A^-) in the acid to form the salt (MA).
- The hydrogen ion (H^+) in the acid together with the hydrogencarbonate ion forming two products – carbon dioxide (CO_2) and water (H_2O).

The general word equations are given below, together with some specific examples.

metal hydrogencarbonate + acid $\rightarrow$ salt + water + carbon dioxide



Such reactions occur in cave formation in many parts of the world, including in the Waitomo cave in New Zealand. Acidic rain with carbonic acid content reacts with limestone rocks to create beautiful caves (**Figure 3**).



Figure 3. Waitomo cave in New Zealand.

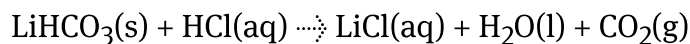
Credit: dabooost, Getty Images

Worked example 4

Complete and balance the equation for the reaction between lithium hydrogencarbonate and hydrochloric acid:



First identify the type of substances involved, then write their formulas and those of the products before balance the equations.



Theory of Knowledge

If you have ever purchased art supplies, you might have noticed the term acid-free on the packaging for paper. Paper is traditionally made from wood pulp containing *lignin*, a fibre that provides strength to the paper, however in time it will break down producing hydrochloric acid. When paper comes in contact with acid, it causes yellow-brown discolouration, known as foxing in the art community. Acid-free paper became commercially available in the 1950s made with lignin removed and infused with calcium hydrogencarbonate or magnesium hydrogencarbonate to neutralise any acids that might come into contact with the paper to prevent foxing.

Paintings and sketches made on traditional paper that have undergone foxing can be restored by neutralising the acids using a base — such as calcium hydroxide or ammonia. Art restoration, however, is a controversial process that risks damaging or altering the piece such that it no longer resembles the original form, making the notion of preservation questionable.

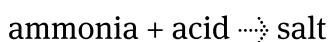
Does knowledge of the chemical process required to save a piece of artwork from damage compel us to use it at the risk of altering the original work?

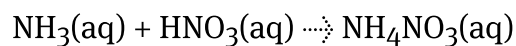
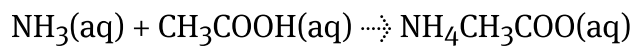
Reaction with ammonia and amines

Acids undergo neutralisation reactions with aqueous ammonia and amines, which are bases, to produce a salt. The reaction involves:

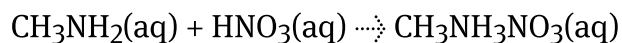
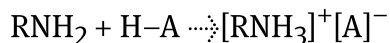
- The nitrogen contains a lone pair that is capable of accepting a hydrogen so the ammonia accepts a hydrogen ion, forming the ammonium ion, NH_4^+ , and the anion from the acid remains unchanged.
- The two oppositely charged species are now attracted to each other electrostatically.

The general word equation is given below, together with some specific examples:





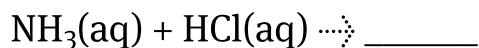
amine + acid $\rightleftharpoons$ salt



Worked example 5

Complete and balance the equations for the reaction between:

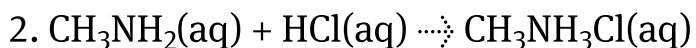
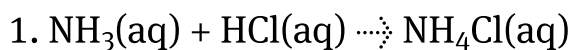
1. ammonia and hydrochloric acid



2. methyl amine and hydrochloric acid



Identify first the type of substances involved, then write their formulas and those of the products, then balance the equations.



Practical skills

Tool 1: Experimental techniques — Applying techniques in separating mixtures

Can you apply the techniques learned in [subtopic S1.1](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-40864/>) to separate the salts formed in neutralisation reactions?

Making connections

Recall from [subtopic R1.1](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43468/>) that reactions can be exothermic or endothermic, and that all energy in reactions is stored in bonds. Neutralisation reactions are exothermic. How can this be explained in terms of bond enthalpies?

Try the simulation in the following activity to check your understanding of the properties of acids and bases.

Activity

- **IB learner profile attribute:** Thinker
- **Approaches to learning:**
 - Thinking skills — Being curious about the natural world; Applying key ideas and facts in new contexts
 - Social skills — Working collaboratively to achieve a common goal
- **Time required to complete activity:** 20 minutes
- **Activity type:** Groups of 2—3

Instructions

1. Open the [online lab room](#)  (<https://web.archive.org/web/20221006082549/http://amrita.olabs.edu.in/olabsub=CHE&cat=PHC&exp=ToStudyThePropertiesOfAcidsNBases&tempId=olabIN>) on properties of acids and bases.
2. Choose the test with sodium carbonate (Solid Na_2CO_3) and Sample 1.
3. Press Help to get instructions on how to run the test.
4. Record your observations.
5. Conclude if the Sample 1 is acidic or basic. Are you correct in your conclusion?
6. Repeat steps 2 to 5 for samples 1 to 4.

7. What would the observations be if samples 1 to 4 were mixed with:

- (a) sodium hydroxide
- (b) ammonia
- (c) sodium hydrogencarbonate?

pH curves for strong acid–base neutralisation reactions

Learning outcomes

By the end of this section you should be able to sketch and interpret the general shape of the pH curve of a strong acid and strong base neutralisation reaction.

Lemon juice, orange juice and grapefruit juice are all acidic. One of the properties of acids is that they have a sour taste. Do all these juices taste as sour as each other? Are they all as acidic as each other? One way to measure the acid content of these juices and other solutions is with titration (**Figure 1**).

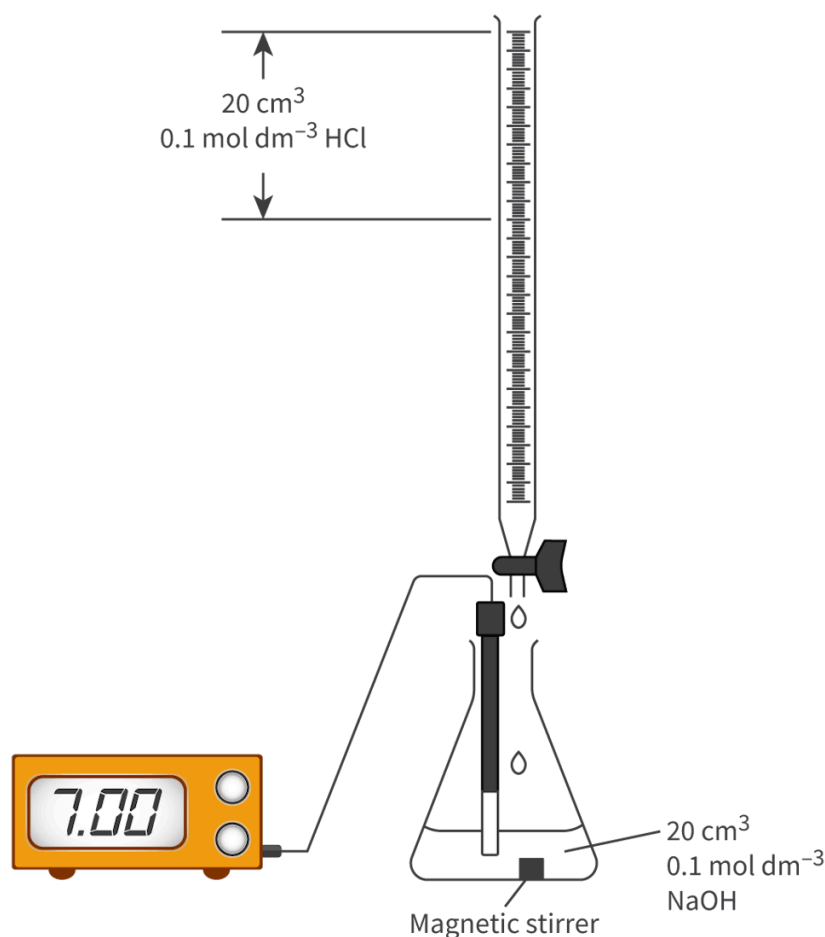


Figure 1. Titration of a strong base (NaOH) with a strong acid (HCl).

 More information for figure 1

The image depicts a titration setup for an experiment involving the titration of a strong base (NaOH) with a strong acid (HCl). The setup includes a tall burette filled with 20 cm³ of 0.1 mol dm⁻³ hydrochloric acid (HCl), indicated by text near the burette. The acid is gradually being released drop by drop into an Erlenmeyer flask below, which contains 20 cm³ of 0.1 mol dm⁻³ sodium hydroxide (NaOH). A magnetic stirrer is placed

under the flask to mix the solution for even distribution of reactants. A pH meter is connected to observe the change in pH level, displaying a reading of 7.00, which suggests the solution is neutral at this point of titration. The setup illustrates the controlled environment required for accurate titration experiments.

[Generated by AI]

Titration

Titration is one of the most widely used techniques in quantitative analysis. An acid–base titration is used to determine the unknown concentration of an acidic or basic solution using a solution of known concentration, which is called a standard solution. **Figure 2** shows the apparatus used in an acid–base titration.

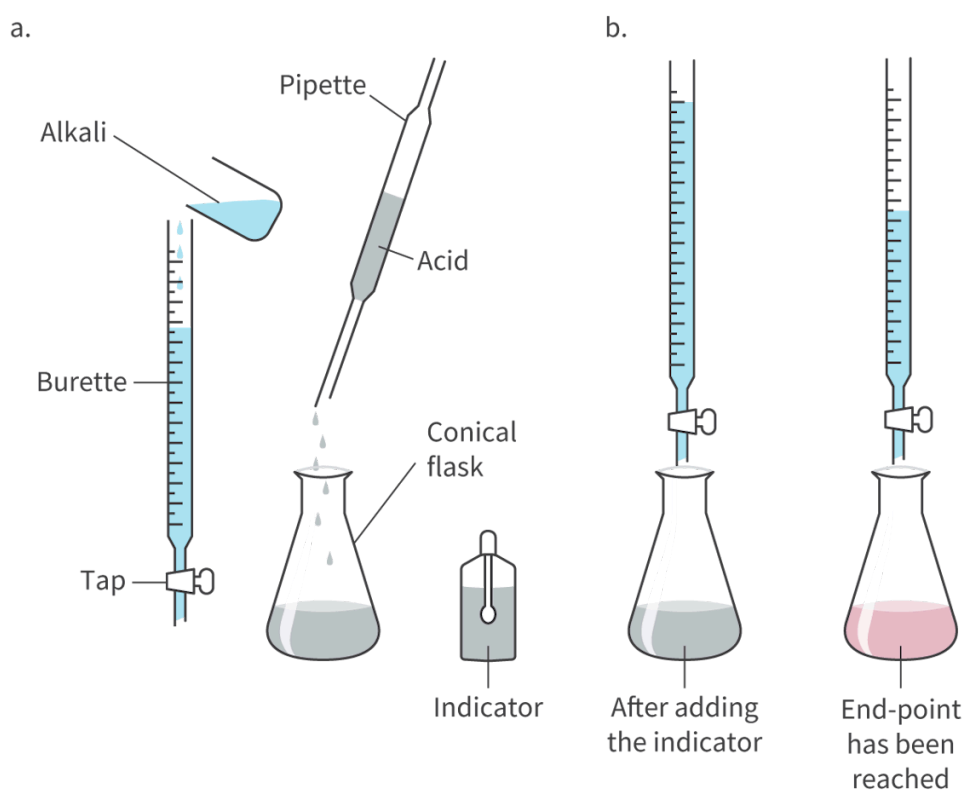


Figure 2. Carrying out a neutralisation by titration.

 More information for figure 2

The image depicts an acid-base titration setup, consisting of two main parts labeled 'a' and 'b.'

In part 'a,' the diagram shows the setup before titration begins: - A burette filled with an alkali solution is positioned vertically on the left side, labeled 'Burette.' The burette has measurement markings, and a tap at the bottom is labeled 'Tap.' - A pipette is held horizontally to transfer an acid solution into a conical flask below. - The conical flask is directly beneath the burette's tip and is labeled. It contains acid and an indicator. - Additional components include a dropper, labeled 'Pipette,' and a diagram of the indicator on the far right, indicating that the color change helps identify the endpoint of the titration.

In part 'b,' the diagram shows two steps of the titration process: 1. On the left, the burette is dispensing the alkali into the conical flask. This step is labeled 'After adding the indicator.' 2. On the right, the titration has reached its endpoint, where the solution in the conical flask has changed color to indicate that the equivalence point has been reached. This is labeled 'End-point has been reached.'

The diagram effectively conveys the procedure and required apparatus for conducting an acid-base titration, highlighting the importance of the visible color change at the endpoint.

[Generated by AI]

The burette is filled with an alkali of known concentration (the titrant). The titrant is added to a measured volume of the acid of unknown concentration (the analyte) until the equivalence point is reached. The equivalence point of the titration is when the acid and base have been reacted in stoichiometrically equivalent amounts. At that point, the solution contains only a salt and water.

The equivalence point is signified when the indicator changes colour. In the example below, the indicator is phenolphthalein. Note that the colour of phenolphthalein in an alkaline solution can be described as pink/purple and can fade to colourless in strong alkali if left to stand. The volume of the standard solution is measured.

Making connections

You studied stoichiometry calculations from subtopic S1.4 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43114/>)? Sometimes reactants are given in excess, sometimes they react completely in the correct ratio, with none in excess. Can you think why is the equivalence point sometimes referred to as the stoichiometric point?

This concept can be applied when testing food components to see if they have too much, too little or just the correct amount of a component.

For example, the content of ascorbic acid in fruit juices in order to compare and determine if the fruit juice has advertised levels of ascorbic acid can be determined by titration. You can explore this [site](https://www.mt.com/in/en/home/applications/laboratory/food-and-beverages/acidity-measurement.html)  (<https://www.mt.com/in/en/home/applications/laboratory/food-and-beverages/acidity-measurement.html>) for more information.

pH curve

In a neutralisation reaction between an acid and a base, there will be a change in pH. This depends on whether the acid is added to the base or vice versa. When a strong acid is added to a strong base, the pH of the solution (as measured by a pH probe) will decrease from a high pH (approximately 14) to a lower pH as the

concentration of H^+ ions increases. The resulting graph showing the change in pH is known as a pH curve. It shows the change in pH vs. the volume of acid or base being added.

In the titration of HCl with NaOH, a student recorded the pH and volume values in **Table 1**

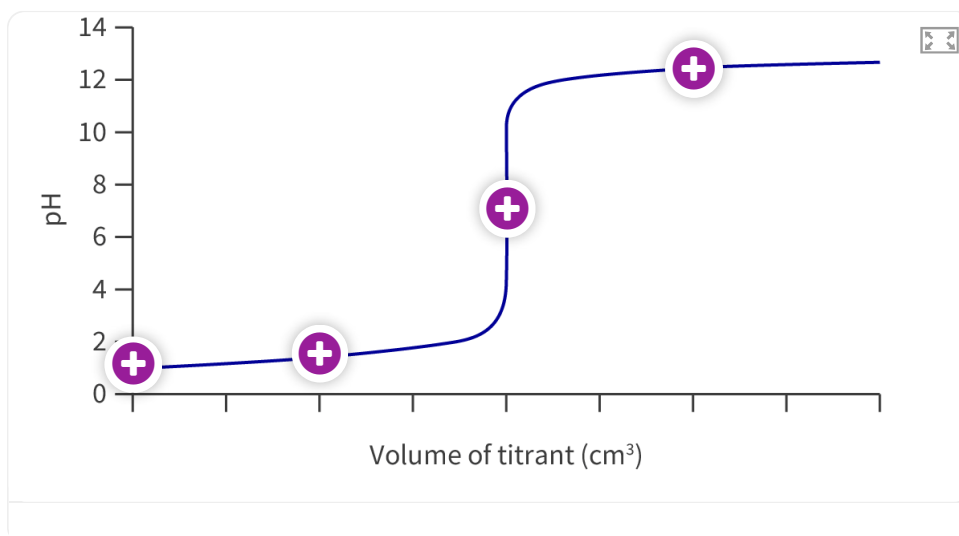
Table 1. Student results for titration of HCl with NaOH

pH	Volume of NaOH added (cm^3)
3.61	0.00
3.92	3.00
4.13	6.00
4.29	9.00
4.43	12.00
4.45	15.00
4.96	18.00
5.00	21.00
5.12	24.00
5.28	27.00
5.46	36.00
5.73	39.00
6.29	42.00
10.84	45.00
11.67	48.00
11.93	51.00

pH	Volume of NaOH added (cm ³)
12.03	54.00
12.18	57.00

Try plotting the data. Can you describe key features of the graph? What can you say about the concentration of acid and base at the equivalence point?

Use **Interactive 1** to investigate the main attributes of a typical pH curve from a titration.



Interactive 1. A Typical pH Curve From a Titration.

 More information for interactive 1

An interactive graph plots pH on the vertical axis, ranging from 0 to 14, and the volume of titrant (cm³) on the horizontal axis. The graph displays an S-shaped curve with four plus icons each representing a hotspot.

The hotspots are named Hotspot 1, Hotspot 2, Hotspot 3, and Hotspot 4 from right to left. The hotspot 1 is located on the curve where the pH is approximately 12. The hotspot 2 is located on the curve where the pH is approximately 7. The hotspot 3 is located on the curve where the pH is approximately 2. The hotspot 4 is located on the curve where the pH is approximately 1.

Clicking on the hotspots reveals the following details:

Hotspot 1:

The curve ends at a high pH (pH of a strong base) since there is now an excess of strong base that ionises completely.

Hotspot 2:

The equivalence point can be determined by finding the point half-way up the steepest part of the curve and drawing a horizontal line across to the y-axis. For the titration of a strong acid and a strong base, the pH at the equivalence point is 7. At the equivalence point, the acid and base have been reacted in stoichiometrically equivalent amounts and the solution contains only salt and water.

Hotspot 3:

With the addition of a strong base to neutralise the acid, the pH gradually increases until it undergoes a sharp increase between pH 3 and pH 11; this is known as the point of inflection.

Hotspot 4:

The pH curve starts at a low pH (pH of a strong acid) since acid ionises completely. This interactive graph helps users understand how the pH changes during a strong acid–strong base titration.

Study skills

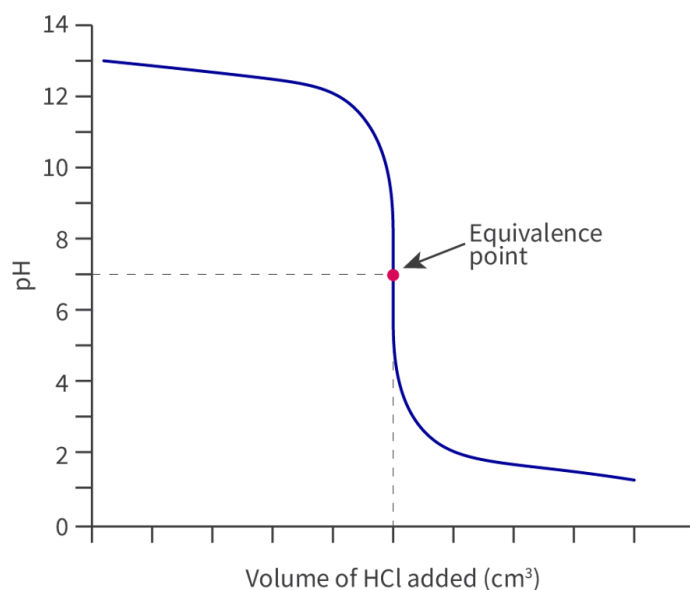
Interpretation of the graph only should include the intercept with the pH axis and equivalence point.

Only monoprotic neutralisation reactions will be assessed.

Worked example 1

Sketch a titration curve for the titration of NaOH with HCl in which the acid is added to the alkali.

Label each section of the graph.



1. The pH curve starts at a high pH (pH of a strong base).
2. With the addition of a strong acid, the pH gradually decreases until it undergoes a sharp decrease between pH 11 and pH 3; this is known as the point of inflection.

3. The equivalence point can be determined by finding the point half-way up the steepest part of the curve and drawing a horizontal line across to the y -axis.
4. For the titration of a strong acid and a strong base, the pH at the equivalence point is 7.
5. At the equivalence point, the acid and base have been reacted in stoichiometrically equivalent amounts and the solution contains only salt and water.
6. The curve ends at a low pH (pH of a strong acid).

Practical skills

Tool 1: Experimental techniques — Applying techniques

Tool 3: Technology — Applying technology to process data

Inquiry 2: Collecting and processing data — Interpreting results, interpret diagrams, graphs and charts

How can titration be used to calculate the concentration of an acid or base in solution?

Think about [subtopic S1.3 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43119/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43119/) and stoichiometry when you perform the acid–base titration in the lab

Remember that when you graph your pH curve, you will use the graph as described above to determine the equivalence point and you will use the information obtained to perform the required calculations.

Activity

- **IB learner profile attribute:** Thinker
- **Approaches to learning:** Thinking skills— Applying key ideas and facts in new contexts
- **Time required to complete activity:** 30 minutes
- **Activity type:** Individual activity

Instructions

You are given the titration results in **Table 1**.

Table 1. Titration results.

Volume of NaOH (cm ³)	
0.00	
2.00	
4.00	
6.00	
8.00	
10.00	
12.00	
14.00	
16.00	
18.00	
20.00	
22.00	
24.00	
1. Use the data to sketch a titration curve. Make sure to clearly label the axis, showing the data points and line connecting them. 2. Label the various parts of the graph describing the stages of the titration. 3. Determine the pH at the equivalence point.	

Ion dissociation constants of weak acids and bases (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to:

- (R3.1.9) Interconvert $[H^+]$, $[OH^-]$, pH and pOH values.
- (R3.1.10) Write K_a and K_b expressions.
- (R3.1.10) Interpret the relative strengths of acids and bases from the K_a , K_b , pK_a or pK_b values.

In [section R3.1.6a](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/strong-and-weak-acids-and-bases-id-44703/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/strong-and-weak-acids-and-bases-id-44703/>), we saw the difference between strong and weak acids and bases. We have seen that the pH of equimolar solutions of weak and strong acid solutions are different, with the weak acid having a higher pH. But how can we quantify this behaviour and how can the pH of a weak acid or base be calculated knowing an equilibrium is established? In [section R3.1.4–5a](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculations-involving-ph-and-[h+]-id-44702/) ([https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculations-involving-ph-and-\[h+\]-id-44702/](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculations-involving-ph-and-[h+]-id-44702/)), you learned about the relationship between H^+ and OH^- , but solutions are typically expressed in terms of pH, which is based on the hydrogen ion concentration. What about the hydroxide concentration? Can we express pOH, just like we express pH?

pOH calculations

We have previously seen that pH indicates the acidity of a solution, and it is related to $[H^+]$ through the logarithmic equation:

$$pH = -\log[H^+]$$

Similarly, to indicate basicity, the pOH of a solution can be calculated by taking the negative log of the hydroxide ion concentration, $[OH^-]$, using the following equation:

Table 1. pOH equation.

Equation	Symbols	Units
$pOH = -\log[OH^-(aq)]$	pOH = a measure of how basic a species is	No units
	$[OH^-(aq)] =$ concentration	Moles per cubic decimetre dm^{-3}

Equation	Symbols	Units
	of aqueous hydroxide ion	

We can also derive the formula for the concentration of hydroxide ions using pOH:

$$[\text{OH}^-] = 10^{-\text{pOH}}$$

Recall from [section R3.1.4–5a](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculations-involving-ph-and-[h+]-id-44702/) ([https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculations-involving-ph-and-\[h+\]-id-44702/](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculations-involving-ph-and-[h+]-id-44702/)), that:

$$[\text{H}^+] \times [\text{OH}^-] = 1 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$$

We can therefore interconvert between $[\text{H}^+]$, $[\text{OH}^-]$, pH and pOH.

Study skills

The equation for pOH is given in the DP chemistry data booklet in [section 1](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/some-relevant-equations-id-45103/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/some-relevant-equations-id-45103/>).

Worked example 1

For a solution with $[\text{OH}^-] = 1.0 \times 10^{-4} \text{ mol dm}^{-3}$ calculate the:

- pOH
- pH

1.

Solution steps	Calculations
Step 1: Write out the values given in the question.	$[\text{OH}^-] = 1.0$
Step 2: Write out the equation.	$\text{pOH} = -\log$
Step 3: Substitute the values given.	$= -\log(1.0)$
Step 4: State the answer with appropriate units and the number of significant figures used in rounding.	$= 4$ Note that pO same reason

2.

Solution steps	Calculations
Step 1: Write out the values given in the question.	$[\text{OH}^-] = 1.0$
Step 2: Write out the required equation.	$[\text{OH}^-] \times [\text{H}^+]$
Step 3: Rearrange the equation.	$[\text{H}^+] = \frac{1 \times}{[\text{O}]}$
Step 4: Substitute the values given.	$= \frac{1 \times 10^{-1}}{1 \times 10^{-4}}$ $= 1 \times 10^{-10}$
Step 5: Write out the required equation.	$\text{pH} = -\log[$
Step 6: Substitute the values calculated.	$\text{pH} = -\log[$
Step 7: State the answer with appropriate units and the number of significant figures used in rounding.	$= 10$

Try these questions for yourself.



Exercises 1 and 2

Click a question to answer



1. True or false?



A solution with $\text{pOH} = 6$ has $[\text{H}^+] = 1 \times 10^{-6} \text{ mol dm}^{-3}$.



2. True or false?



A solution with $[\text{H}^+] = 1 \times 10^{-4} \text{ mol dm}^{-3}$ has $\text{pOH} = 10$.

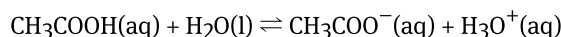


Check answers

Higher level (HL)

Acid and base dissociation constants

Strong acids and alkalis completely dissociate (or ionise) into ions in aqueous solution and as such the pH of solutions of strong acids and bases is relatively straightforward to calculate as seen in [section R3.1.4-5a](#) ([https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculations-involving-ph-and-\[h+\]-id-44702/](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculations-involving-ph-and-[h+]-id-44702/)). However, as seen in [section R3.1.6a](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/strong-and-weak-acids-and-bases-id-44703/>), there are a wide range of acids and bases where the degree of dissociation or ionisation is not complete. Such acids and bases are referred to as weak acids and bases - there exists an equilibrium between the dissociated and undissociated forms. The following equation shows the equilibrium for a solution of ethanoic acid, CH₃COOH.

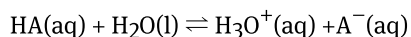


Weak acids such as ethanoic acid partially dissociate in solution. Unlike strong acids and strong bases, we cannot determine the [H⁺] or [OH⁻] from their initial concentrations. This is because each weak acid or weak base has a different degree of dissociation.

Just like equilibrium reactions, the dissociation of a weak acid or weak base has a constant associated with it as seen in [section R2.3.2](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-equilibrium-constant-kc-and-magnitude-of-the-kc-id-46067/>), which tells us the extent of this dissociation and where the position of equilibrium lies as seen in [section R2.3.3](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-reaction-quotient-q-hl-id-46068/>). For weak acids, this is the acid dissociation constant, K_a .

The acid dissociation constant, K_a

The dissociation of a generic weak acid, HA, can be represented as:



The expression for the acid dissociation constant, K_a , for the dissociation reaction is given in **Tables 2a** and **2b**.

Table 2a. Acid dissociation constant.

Equation	Symbols	Units
$K_a = \frac{[\text{H}_3\text{O}^+][\text{A}^-]}{[\text{HA}]}$	K_a is the acid dissociation constant	Moles per cubic deci (mol dm ⁻³)
	[HA] is the concentration of the acid.	Moles per cubic deci (mol dm ⁻³)
	[H ₃ O ⁺] is the concentration of the hydronium ion.	Moles per cubic deci (mol dm ⁻³)

Equation	Symbols	Units
	$[A^-]$ is the concentration of the A^- ion.	Moles per cubic decimetre (mol dm ⁻³)

or

Table 2b. Acid dissociation constant

Equation	Symbols	Units
$K_a = \frac{[H^+][A^-]}{[HA]}$	K_a is the acid dissociation constant	Moles per cubic decimetre (mol dm ⁻³)
	$[HA]$ is the concentration of the acid.	Moles per cubic decimetre (mol dm ⁻³)
	$[H^+]$ is the concentration of the hydrogen ion.	Moles per cubic decimetre (mol dm ⁻³)
	$[A^-]$ is the concentration of the A^- ion.	Moles per cubic decimetre (mol dm ⁻³)

Note that this reaction equation is an example of heterogeneous equilibrium because there are a mixture of states - both liquid and aqueous - in the reaction. Concentration is used to quantify mixtures, so the aqueous species are included in the equilibrium expression. However, water, is a pure substance and is therefore not expressed by concentration. Therefore, water is not included in the final equilibrium expression.

K_a is known as the acid dissociation constant (or acid ionisation constant). It has a fixed value for a particular acid at a particular temperature - it is temperature-dependent.

Worked example 2

Write the K_a expression for the dissociation of the weak acid, HF.

Solution steps	Calculations
Step 1: Write out the dissociation equation.	$HF(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + F^-(aq)$
Step 2: Write the K_a expression,	$K_a = \frac{[H_3O^+][F^-]}{[HF]}$

How do you think the K_a value and the strength of an acid are related?

As seen in subtopic R2.3 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43482/>), the higher the extent of the reaction, the more favoured the products are. Therefore, the higher the concentrations of the products. For a weak acid, the reactants are favoured. The weaker the acid, the less it dissociates, and the lower the concentration the products and the higher the concentration of the acid, making the K_a ratio small. As such, the value of the K_a tells us about the extent of dissociation and therefore the strength of a weak acid and therefore the degree of dissociation. A weak acid with a higher K_a value is stronger and more dissociated in solution than a weak acid with a lower K_a value (**Figure 1**).

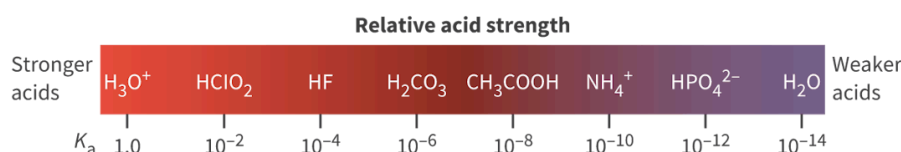


Figure 1. K_a values.

 More information for figure 1

This image is a diagram illustrating the relative acid strength using K_a values. The diagram is a horizontal bar with gradients transitioning from red to purple, indicating stronger to weaker acids. At the leftmost, the bar indicates 'Stronger acids' with a label for a K_a of 1.0 showing H_3O^+ . Progressing to the right, the K_a values decrease: $HClO_2$ (10^{-2}), HF (10^{-4}), H_2CO_3 (10^{-6}), CH_3COOH (10^{-8}), NH_4^+ (10^{-10}), HPO_4^{2-} (10^{-12}), and ending with H_2O at a K_a of 10^{-14} , which is the weakest. Above the bar, the text 'Relative acid strength' is written, providing context for the strength gradient depicted in the diagram.

[Generated by AI]

Worked example 3

Which acid is strongest? Which is the weakest?

Acid	Formula	K_a
Methanoic acid	$HCOOH$	1.8×10^{-4}
Ethanoic acid	CH_3COOH	1.8×10^{-5}
Propanoic acid	C_2H_5COOH	1.4×10^{-5}

The largest K_a value is 1.8×10^{-4} and therefore the strongest acid is methanoic acid.

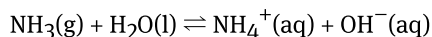
The lowest K_a value is 1.4×10^{-5} and therefore the weakest acid is propanoic acid.

Study skills

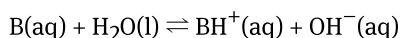
In section R1.4.4a (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/gibbs-energy-and-equilibrium-hl-id-44767/>), we looked at the relationship between the equilibrium position and the Gibbs free energy change, ΔG . Students should be aware that the ΔG can also be calculated using the K_a or K_b for a weak acid or base (and vice versa) instead of the equilibrium constant, K_c .

The base dissociation constant, K_b

Ammonia, NH_3 , is an example of a weak base that partially dissociates in solution:



In a similar argument to that used above for weak acids, we can consider the ionisation of a weak base using the generic weak base B:



and derive an expression for K_b , the base dissociation constant.

Table 3. Base dissociated constant.

Equation	Symbols	Units
$K_b = \frac{[\text{BH}^+][\text{OH}^-]}{[\text{B}]}$	K_b is the base dissociation constant.	Moles per cubic deci (mol dm ⁻³)
	$[\text{B}]$ is the concentration of the base.	Moles per cubic deci (mol dm ⁻³)
	$[\text{BH}^+]$ is the concentration of the BH^+ ion.	Moles per cubic deci (mol dm ⁻³)
	$[\text{OH}^-]$ is the concentration of the hydroxide ion.	Moles per cubic deci (mol dm ⁻³)

K_b , the base dissociation constant, has a fixed value for a particular base at a specific temperature.

Worked example 4

Write the K_b expression for the dissociation of the weak base, CH_3NH_2 .

Solution steps	Calculations
Step 1: Write out the dissociation equation.	$\text{CH}_3\text{NH}_2(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{CH}_3\text{NH}_3^+(\text{aq}) + \text{OH}^-(\text{aq})$

Solution steps	Calculations
Step 2: Write the K_b expression.	$K_b = \frac{[\text{CH}_3\text{NH}_3^+][\text{OH}^-]}{[\text{CH}_3\text{NH}_2]}$

As seen in subtopic R2.3 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43482/>), the higher the extent of the reaction, the more favoured the products are. Therefore, the higher the concentrations of the products. For a weak base, the reactants are favoured. The weaker the base, the less it dissociates, and the lower the concentration of the products and the higher the concentration of the base, making the K_b ratio small. As such, Similar to the K_a , the value of K_b tells us about the strength of the base and therefore the degree of dissociation. A weak base with a high K_b value is stronger and more dissociated than a weak base with a low K_b value (Figure 2).

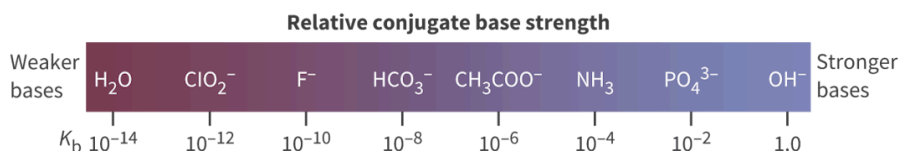


Figure 2. K_b values.

 More information for figure 2

The image is a diagram illustrating the relative conjugate base strength. It shows a gradient bar with the label 'Relative conjugate base strength' at the top. On the left, it mentions 'Weaker bases', and on the right, 'Stronger bases'. The diagram lists various bases along the bar, each associated with a specific K_b value, which indicates their strength. From left to right:

- H_2O with $K_b 10^{-14}$
- ClO_2^- with $K_b 10^{-12}$
- F^- with $K_b 10^{-10}$
- HCO_3^- with $K_b 10^{-8}$
- CH_3COO^- with $K_b 10^{-6}$
- NH_3 with $K_b 10^{-4}$
- PO_4^{3-} with $K_b 10^{-2}$
- OH^- with $K_b 1.0$

[Generated by AI]

Worked example 5

Which base is strongest and which is the weakest?

Base	K_b
A	1.8×10^{-4}
B	1.8×10^{-3}
C	1.4×10^{-6}

The largest K_b value is $1.8 \times 10^{-3} \text{ mol dm}^{-3}$ and therefore the strongest base is B.

The lowest K_b value is $1.4 \times 10^{-6} \text{ mol dm}^{-3}$ and therefore the weakest base is C.

Study skills

The values of K_a and K_b are constant for a specific acid and base respectively, at a specified temperature. K_a and K_b values tell us about the strength of a weak acid or weak base and the degree of dissociation.

These values will be given in the exam questions where needed.

pK_a and pK_b

Like we calculated pH from $[H^+]$ and pOH from $[OH^-]$ in order to convert numbers with exponents to easier numbers for comparison purposes, we can calculate pK_a from K_a and pK_b from K_b . Here are the formulas that you can use:

$$pK_a = -\log K_a \quad \text{and} \quad pK_b = -\log K_b$$

$$K_a = 10^{-pK_a} \quad \text{and} \quad K_b = 10^{-pK_b}$$

Look at the K_a , pK_a , K_b and pK_b values in **Table 4** and **Table 5**.

Can you find a relationship between strength of acid, K_a and pK_a , and between strength of base, K_b and pK_b ?

Table 4. The K_a and pK_a values of some weak acids.

Acid	Formula	K_a	
Methanoic acid	HCOOH	1.8×10^{-4}	
Ethanoic acid	CH ₃ COOH	1.8×10^{-5}	
Propanoic acid	C ₂ H ₅ COOH	1.4×10^{-5}	

Table 5. The K_b and pK_b values of some weak bases.

Base	Formula	K_b	p
Ammonia	NH_3	1.8×10^{-5}	4
Methylamine	CH_3NH_2	4.6×10^{-4}	3
Ethylamine	$\text{C}_2\text{H}_5\text{NH}_2$	4.5×10^{-4}	3

As the values above show, the stronger the acid, the larger the K_a and the smaller the pK_a . Similarly, the stronger the base, the larger the K_b and the smaller the pK_b (**Figure 3**).

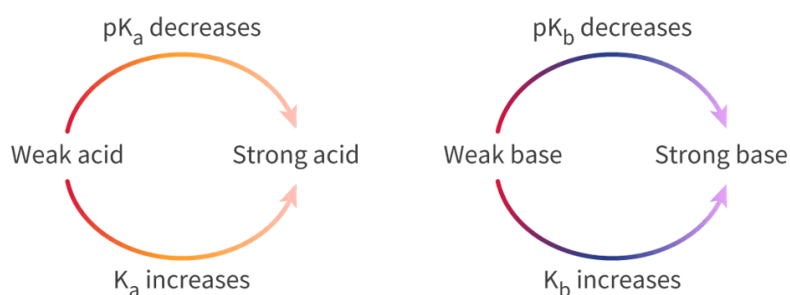


Figure 3. The relationship between K_a and pK_a with acid strength, and the equivalent relationship for bases.

 More information for figure 3

The image is a diagram with two circular arrows. The left circle represents acids and includes the sequence: 'Weak acid' at the bottom left, 'Ka increases' in clockwise direction at the bottom, 'Strong acid' at the top right, and 'pKa decreases' at the top. The right circle represents bases and includes the sequence: 'Weak base' at the bottom left, 'Kb increases' in clockwise direction at the bottom, 'Strong base' at the top right, and 'pKb decreases' at the top. The diagram visually demonstrates that as the strength of an acid increases, its K_a value increases and pK_a value decreases, and similarly for bases with K_b and pK_b .

[Generated by AI]

Worked example 6

The pK_b of a base is 3.34.

Calculate its K_b .

Solution steps

Step 1: Write out the values given in the question.

Solution steps

Step 2: Write out the equation.

Step 3: Substitute the values given.

Step 4: State the answer with appropriate units and the number figures used in rounding.

Worked example 7

The K_a of an acid is 1.8×10^{-5} .

Calculate its pK_a .

Solution steps

Step 1: Write out the values given in the question.

Step 2: Write out the equation.

Step 3: Substitute the values given.

Step 4: State the answer with appropriate units and the number significant figures used in rounding.

Worked example 8

Looking at the values of K_a , pK_a , K_b and pK_b in **Table 4** above.

Which acid is stronger, methanoic acid or ethanoic acid?

Since the pK_a of methanoic acid is less than the pK_a of ethanoic acid, methanoic acid is stronger.

Worked example 9

Looking at the values of K_a , pK_a , K_b and pK_b in the **Table 5** above.

Which base is weaker, ammonia or methylamine?

Since the pK_b of methylamine is lower than the pK_b of ammonia,

ammonia is a weaker base than methylamine.

Use **Interactive 1** to consolidate your understanding of K_a and pK_a .

Drag the words into the correct boxes

1. The pK_a of an acid is 4, therefore the K_a is .
2. If an acid A has a pK_a value of 4, and an acid B a pK_a of 10, then A is than B.
3. If a base A has a pK_b value of 6 and a base B a pK_b of 2, then base A is than B.

1×10^{-4}

weaker

stronger

✓ Check

Interactive 1. The strength of some weak acids and bases.

Activity

- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 20 minutes
- **Activity type:** Individual activity

Instructions:

1. Go on a hunt! Find the K_a or K_b of at least eight materials found in real contexts.
2. Calculate their pK_a or pK_b .
3. Create a rating scale, arranging the materials by acid—base strength. Include the substance and its K_a or K_b and pK_a or pK_b .

K_a and K_b calculations (HL)

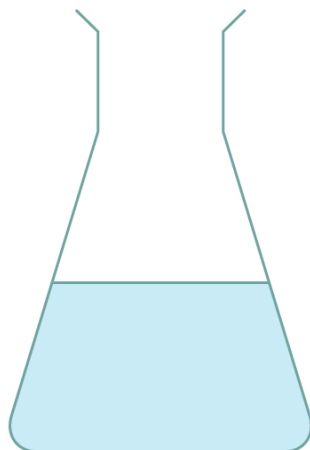
Higher level (HL)

Learning outcomes

By the end of this section you should be able to solve problems involving K_a and K_b values.

Use **Interactive 1** to remind yourself about K_a and pH.

Look at the relationship between K_a and pH.



0.10 M
hydrochloric acid
HCl
100% ionised
 $K_a = 1000$

↺ Turn

Card 1 of 3



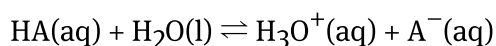
Interactive 1. Relating K_a and pH.

Now that you understand the difference between strong and weak acids, can you calculate the pH of a weak acid in the same way in which we calculated pH of a strong acid? Why? What is different?

We will revisit relationships that we have studied throughout this subtopic and put them in mathematical relationships. We will start with associating numbers to the relative strength of acids and bases in conjugate acid–base pairs.

Conjugate acid—base pairs

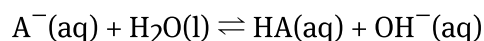
For an acid:



and

$$K_a = \frac{[\text{H}_3\text{O}^+][\text{A}^-]}{[\text{HA}]}$$

For its conjugate base A[−]:



and

$$K_b = \frac{[\text{OH}^-][\text{HA}]}{[\text{A}^-]}$$

If K_a and K_b are multiplied then:

$$K_a \times K_b = \frac{[\text{H}_3\text{O}^+][\text{A}^-]}{[\text{HA}]} \times \frac{[\text{OH}^-][\text{HA}]}{[\text{A}^-]}$$

which simplifies to:

$$K_a \times K_b = [\text{H}_3\text{O}^+][\text{OH}^-]$$

But we know from section R3.1.4–5a ([https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculations-involving-ph-and-\[h+\]-id-44702/](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculations-involving-ph-and-[h+]-id-44702/)) that:

$$[\text{H}_3\text{O}^+][\text{OH}^-] = K_w = 1 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$$

Therefore:

$$K_a \times K_b = 1 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$$

Taking the logarithm of both sides, we obtain:

$$\text{p}K_a + \text{p}K_b = 14$$

Therefore, for a conjugate acid–base pair:

- $K_a \times K_b = 1 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$
- $\text{p}K_a + \text{p}K_b = 14$

We can conclude that the stronger the acid and therefore the higher its K_a value, then the lower the K_b value of the conjugate base, and the weaker the conjugate base.

Study skills

Remember for any conjugate acid–base pair:

- $K_a \times K_b = K_w$
- $\text{p}K_a + \text{p}K_b = \text{p}K_w$
- $\text{p}K_a + \text{p}K_b = 14.00$ at 298 K

Calculations of pH for weak acids and bases

When calculating the pH of strong acids and bases we assume full dissociation into ions in aqueous solution. Because of this, we can use the initial concentration of the acid or base to calculate the pH or pOH as seen in [section R3.1.4–5a](#)

([https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculations-involving-ph-and-\[h+\]-id-44702/](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculations-involving-ph-and-[h+]-id-44702/)).

Weak acids and weak bases only partially dissociate in solution, which means that unlike for strong acids and bases, we cannot determine the $[H^+]$ or $[OH^-]$ in solution from their initial concentrations.

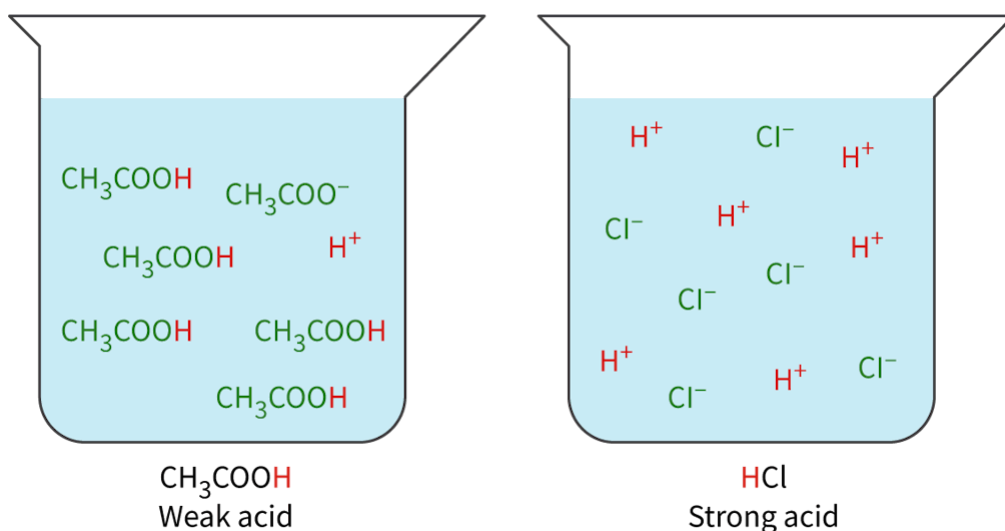


Figure 1. Dissociation of weak vs. strong acid.

 More information for figure 1

The image shows two beakers, each containing a liquid with dissociated molecules. The left beaker is labeled with the chemical notation for weak acid, acetic acid (CH_3COOH). Inside the left beaker, several undissociated acetic acid molecules (CH_3COOH) are present alongside a few dissociated ions, acetate (CH_3COO^-) and hydrogen (H^+). This illustrates partial dissociation characteristic of weak acids.

The right beaker is labeled as containing a strong acid, hydrochloric acid (HCl). In this beaker, ions fully dissociated into hydrogen (H^+) and chloride (Cl^-) are depicted, indicating complete dissociation characteristic of strong acids. This comparison visually demonstrates the difference between the dissociation of weak and strong acids in aqueous solutions.

[Generated by AI]

As seen in **Figure 1**, for weak acids, only a few molecules dissociate. The number of dissociated molecules is very small. Therefore, if the acid is weak, the equilibrium concentration of the undissociated acid molecules can be taken as the initial concentration of the acid, $[HA]_{eq} = [HA]_{initial}$. The assumption here is that very few

acid molecules have actually dissociated. A similar assumption is made for the concentration of the undissociated molecules of a weak base at equilibrium; namely, that the equilibrium concentration of the weak base is approximately equal to the initial concentration of the weak base, $[B]_{\text{eq}} = [B]_{\text{initial}}$.

Worked example 1

A $0.0100 \text{ mol dm}^{-3}$ solution of a weak acid HA at 298 K has a pH of 4.17. Calculate the K_a of the acid.

First, determine the $[H^+]$ from pH.

$$\begin{aligned} [H^+] &= 10^{-\text{pH}} \\ &= 10^{-4.17} \\ &= 6.76 \times 10^{-5} \text{ mol dm}^{-3} \end{aligned}$$

The K_a equation is:

$$K_a = \frac{[H^+][A^-]}{[HA]}$$

Let x be $[H^+]$ at equilibrium.

Using an ICE table gives:

	HA (aq) (mol dm^{-3})	H^+ (ac) (mol dm^{-3})
Initial concentration	0.0100	0
Change in concentration	$-x$	x
Equilibrium concentration	$0.0100 - 6.76 \times 10^{-5}$	$6.76 \times 10^{-5} \text{ dm}^{-3}$

Note that the ratio of $[H^+]$ to $[A^-]$ is 1:1, therefore:

$$\begin{aligned} x &= [H^+] \\ &= [A^-] \\ &= 6.76 \times 10^{-5} \text{ mol dm}^{-3} \end{aligned}$$

and

$$[HA] = 0.0100 - 6.76 \times 10^{-5}$$

When rounding to 4 significant figures:

$$0.0100 - 6.76 \times 10^{-5} \cong 0.0100 \text{ mol dm}^{-3}$$

therefore we can omit the $-x$ term in the calculations below. This is an assumption and approximation that can almost always be applied in this course especially because K_a values are relatively small and therefore x is a small value.

Substitute the values into the K_a equation:

$$\begin{aligned} K_a &= \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]} \\ &= \frac{(6.76 \times 10^{-5})^2}{0.01} \\ &= 4.60 \times 10^{-7} \text{ mol dm}^{-3} \end{aligned}$$

Worked example 2

The value of the K_a for ethanoic acid, CH_3COOH , is 1.8×10^{-5} at 25°C . Calculate the pH of a 0.10 mol dm^{-3} solution of CH_3COOH .

Ethanoic acid dissociates according to the equation:



The K_a equation is:

$$K_a = \frac{[\text{CH}_3\text{COO}^-][\text{H}^+]}{[\text{CH}_3\text{COOH}]}$$

Let x be $[\text{H}^+]$ at equilibrium.

Using an ICE table gives:

	$\text{CH}_3\text{COOH}(\text{aq})$ (mol dm^{-3})	$\text{H}^+(\text{aq})$ (mol dm^{-3})
Initial concentration	0.10	0
Change in concentration	$-x$	x
Equilibrium concentration	$0.10 - x$	x

Remember the assumption:

Because of the very small value of the K_a , we can make the assumption that $0.10 - x \approx 0.10$

For this acid $[H^+] = [CH_3COO^-] = x$, therefore:

$$K_a = \frac{[CH_3COO^-][H^+]}{[CH_3COOH]}$$

$$1.8 \times 10^{-5} = \frac{x^2}{0.10}$$

$$\begin{aligned} x^2 &= 1.8 \times 10^{-5} \times 0.10 \\ &= 1.8 \times 10^{-6} \end{aligned}$$

$$x = 1.3 \times 10^{-3}$$

$$[H^+] = 1.3 \times 10^{-3} \text{ mol dm}^{-3}$$

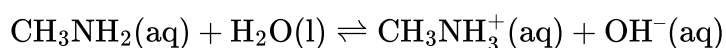
$$\text{pH} = -\log[H^+]$$

$$\begin{aligned} \text{pH} &= -\log(1.3 \times 10^{-3}) \\ &= 2.89 \end{aligned}$$

Worked example 3

The value of the K_b for methylamine (IUPAC name methanamine), CH_3NH_2 , is 4.6×10^{-4} at 25°C. Calculate the pH of a 0.10 mol dm^{-3} solution of CH_3NH_2 .

Methylamine reacts with water as shown in the equation:



The K_b equation is:

$$K_b = \frac{[CH_3NH_3^+][OH^-]}{[CH_3NH_2]}$$

Let x be $[H^+]$ at equilibrium.

Because of the very small value of the K_b , we can make the assumption that the $[CH_3NH_2]_{eq} = [CH_3NH_2]_{initial}$.

Using an ICE table gives:

	CH ₃ NH ₂ (aq) (mol dm ⁻³)	OH ⁻ (mol dm ⁻³)
[] initial (mol dm ⁻³)	0.10	0
Change in [] (mol dm ⁻³)	-x	x
[] eq (mol dm ⁻³)	0.10 - x	x

For this base:

$$[\text{OH}^-] = [\text{CH}_3\text{NH}_3^+] \\ = x$$

Because of the weak nature of the base, K_b is small, and we can make the assumption that $0.10 - x \approx 0.10$, therefore:

$$4.6 \times 10^{-4} = \frac{x^2}{0.10}$$

$$x^2 = 4.6 \times 10^{-4} \times 0.10$$

$$x = 6.8 \times 10^{-3}$$

$$[\text{OH}^-] = 6.8 \times 10^{-3} \text{ mol dm}^{-3}$$

$$\text{pOH} = -\log[\text{OH}^-]$$

$$\text{pOH} = -\log(6.8 \times 10^{-3})$$

$$\text{pOH} = 2.17$$

At 25 °C,

$$\text{pH} + \text{pOH} = 14.00$$

$$\text{pH} = 14.00 - 2.17 \\ = 11.83$$

Making connections

How can we simplify calculations when equilibrium constants K_a and K_b are very small? (see [subtopic R2.3 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43482/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43482/))

Activity

- **IB learner profile attribute:** Knowledgeable:
- **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 20 minutes
- **Activity type:** Individual activity

Instructions

1. Research the K_a of at least six acids then the K_b of six bases.
2. Create a spreadsheet like the one below.
3. Calculate the K_b of the conjugate base for the acids, and the K_a of conjugate acid for the bases.
4. Calculate the corresponding pK_a and the pK_b .

Acid/ base	K_a/K_b value	K_b/K_a of conjugate base/acid

pH of salt solutions (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to:

- construct equations for the hydrolysis of ions in a salt
- predict the effect of each ion on the pH of the salt solution.



Figure 1. Sodium hydrogencarbonate relieves heartburn caused by acid reflux.

Credit: Juanmonino, Getty Images

The salt sodium hydrogencarbonate is one of the ingredients in tablets that provide relief from heartburn. Can all salts be used to relieve heartburn? The salt sodium chloride, for example, does not give relief from heartburn. Why? In this section, you will learn that salts can be acidic, basic or neutral.

Salt hydrolysis

Neutralisation reactions were covered in [section R3.1.7](#)

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/reactions-of-acids-id-46056/>). Although salts are the products of a neutralisation reaction, they do not all form neutral solutions. Salt solutions can be acidic or basic too.

Salts of strong acids and strong bases

In [section S2.1.3b](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/physical-properties-of-ionic-compounds-id-43559/>), you learned that ionic compounds will dissociate in water to form ions (**Figure 2**).

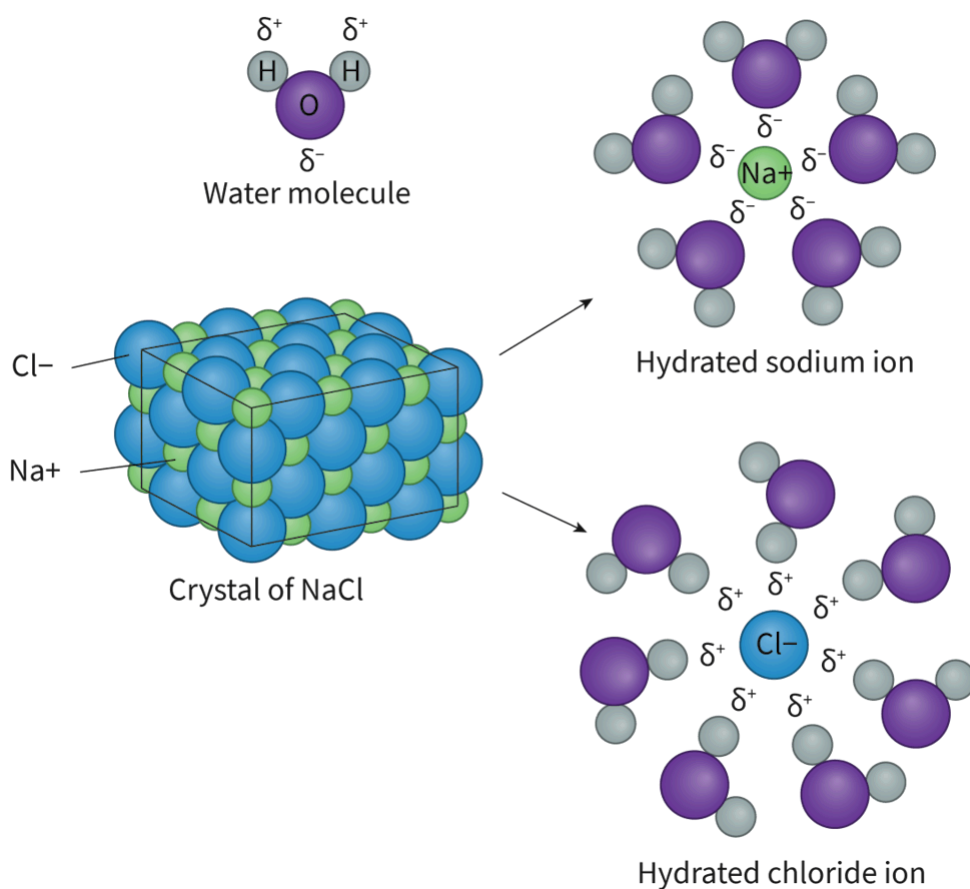


Figure 2. Dissolution of an ionic compound breaking apart the ions.

 More information for figure 2

The diagram illustrates the process of sodium chloride (NaCl) dissociating in water to form hydrated ions. At the center left, there is a crystal structure of NaCl, depicted as a 3D lattice of alternating green spheres representing sodium ions (Na⁺) and blue spheres representing chloride ions (Cl⁻). Arrows point to two separate hydrated ions formations, showing the dissociation process.

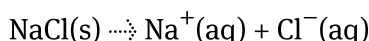
At the top right, a sodium ion (Na⁺) is centrally surrounded by water molecules, depicted with purple and grey colors. Each water molecule has a partial negative charge (δ^-) towards the sodium ion and a partial positive

charge ($\delta+$) on the hydrogen atoms.

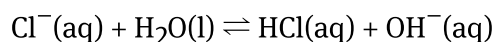
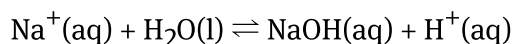
Below, a chloride ion (Cl^-) is surrounded by water molecules in a similar hydration shell, with the hydrogen end of each water molecule facing the chloride ion due to the attraction of opposite charges. The spheres surrounding both ions represent the hydration shells formed by the partial positive and negative charges of the water molecules.

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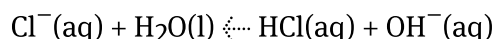
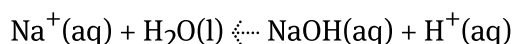
When a strong acid and strong base react, the resulting salt consists of ions that are neither acidic nor basic, which means that neither is able to hydrolyse water. The solution will, therefore, have a pH of 7 at 298 K. Examples of such salts are sodium chloride (NaCl), potassium nitrate (KNO_3) and sodium sulfate (Na_2SO_4). They dissociate to give neutral ions and thus a solution with a pH of 7.



When sodium ions and chloride ions react with water, think about what possible products could form. Using ionisation equations, sodium cations could form a bond with hydroxide anions and chloride anions could form a bond with hydrogen cations, producing H^+ ions and OH^- ions, in their respective equations.



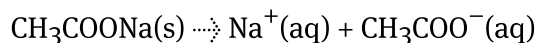
Recall from [section R3.1.6a \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/strong-and-weak-acids-and-bases-id-44703/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/strong-and-weak-acids-and-bases-id-44703/), that sodium hydroxide is a strong base and hydrochloric acid is a strong acid. Strong acids and bases fully ionise in water, so therefore, you will not find these products in solution.



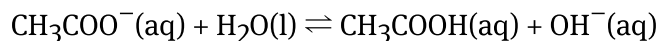
Therefore, NaCl does not have any acid–base activity and will have a pH of 7 in solution.

Salts of weak acids and strong bases

An example of hydrolysis by the salt formed from a weak acid and strong base is that of sodium ethanoate (CH_3COONa). This salt fully dissociates in solution to form sodium ions and ethanoate ions:

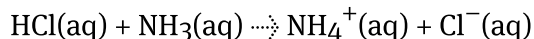


As explained above, the sodium ion does not have acid-base activity. The ethanoate ion (CH_3COO^-) is the conjugate base reacts with water to form ethanoic acid, a weak acid, which does not fully ionise in water. This reaction will therefore show equilibrium, forming OH^- ions that make the solution alkaline, with a pH above 7 at 298 K.

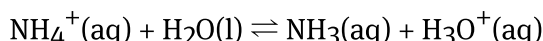


Salts of strong acids and weak bases

In the reaction of a strong acid and a weak base, the conjugate acid of the weak base is strong enough to hydrolyse water. For example, the reaction of hydrochloric acid and ammonia produces the ammonium ion according to the following reaction:



As shown above, the chloride ion does not have acid–base activity. The ammonium ion reacts with water to form ammonia, a weak base, which does not fully ionise in water. This reaction will show equilibrium, forming H_3O^+ ions that make the solution acidic, with a pH below 7 at 298 K



Salts of weak acids and weak bases

What do you think would happen to the solution if both of the ions in the salt showed acid–base activity?

The pH of a salt formed by the reaction of a weak acid and a weak base depends on the K_a and K_b values of the acids and bases involved. If the parent acid and base have approximately the same value, neither ion will hydrolyse and the salt produced will have a pH of around 7.0. Use **Interactive 1** to complete a summary of the pH of different salts produced in neutralisation reactions.

Reaction	Example	Product	Hydrolysis	Acidic, basic or neutral	pH at 298 K
Strong acid + strong base	hydrochloric acid + sodium hydroxide				
Weak acid + strong base	ethanoic acid + sodium hydroxide				
Strong acid + weak base	hydrochloric acid + ammonia				
Weak acid + weak base	ethanoic acid + ammonia				

		ammonium ion hydrolyses water	both hydrolyse water	acidic	cannot generalise
ammonium ethanoate	ammonium chloride	neither ion hydrolyses water	ethanoate ion hydrolyses water	basic	
				depends on strength of conjugate acid or base	
sodium chloride	sodium ethanoate			neutral	> 7.0

☒ Check

Interactive 1. Summary table of the acidic and basic behaviour of salts.

Study skills

Examples should include the ammonium ion NH_4^+ , the carboxylate ion RCOO^- , the carbonate ion CO_3^{2-} , and the hydrogencarbonate ion HCO_3^- .

The acidity of hydrated transition element ions and (aq) is not required.

Worked example 1

Predict if the pH is 7, less than 7, greater than 7 for the following salts in aqueous solution:

1. Ammonium chloride
2. Potassium ethanoate
3. Sodium hydrogencarbonate
4. Lithium carbonate
5. Lithium chloride

1.

Solution steps	Calculations
Step 1: Write dissociation equation.	$\text{NH}_4\text{Cl}(\text{s}) \rightarrow \text{NH}_4^+(\text{aq}) + \text{Cl}^-(\text{aq})$
Step 2: Identify which ion undergoes hydrolysis.	Ammonium ions hydrolyse
Step 3: Write the equation.	$\text{NH}_4^+(\text{aq}) + \text{H}_2\text{O} \rightleftharpoons \text{NH}_3(\text{aq}) + \text{H}_3\text{O}^+(\text{aq})$
Step 4: State the answer	The presence of hydronium ions means the pH is less than 7.

2.

Solution steps	Calculations
Step 1: Write dissociation equation.	$\text{CH}_3\text{COOK}(\text{s}) \rightarrow \text{K}^+(\text{aq}) + \text{CH}_3\text{COO}^-(\text{aq})$
Step 2: Identify which ion undergoes hydrolysis.	Ethanoate ions hydrolyse water.

Solution steps	Calculations
Step 3: Write the equation.	$\text{CH}_3\text{COO}^-(\text{aq}) + \text{H}_2\text{O} \rightleftharpoons \text{CH}_3\text{COOH}(\text{aq}) + \text{OH}^-(\text{aq})$
Step 4: State the answer	The presence of hydroxide ions makes the pH is more than 7.

3.

Solution steps	Calculations
Step 1: Write dissociation equation.	$\text{NaHCO}_3(\text{s}) \rightarrow \text{Na}^+(\text{aq}) + \text{HCO}_3^-(\text{aq})$
Step 2: Identify which ion undergoes hydrolysis.	Hydrogencarbonate ions hydrolyse water.
Step 3: Write the equation.	$\text{HCO}_3^-(\text{aq}) + \text{H}_2\text{O} \rightleftharpoons \text{H}_2\text{CO}_3(\text{aq}) + \text{OH}^-(\text{aq})$
Step 4: State the answer	The presence of hydroxide ions therefore the pH is more than 7.

4.

Solution steps	Calculations
Step 1: Write dissociation equation.	$\text{Li}_2\text{CO}_3(\text{s}) \rightarrow 2\text{Li}^+(\text{aq}) + \text{CO}_3^{2-}(\text{aq})$
Step 2: Identify which ion undergoes hydrolysis.	Carbonate ions hydrolyse water.
Step 3: Write the equation.	$\text{CO}_3^{2-}(\text{aq}) + \text{H}_2\text{O} \rightleftharpoons \text{HCO}_3^-(\text{aq}) + \text{OH}^-(\text{aq})$

Solution steps	Calculations
Step 4: State the answer	The presence of hydroxide ions make the pH is more than 7.

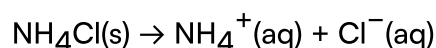
5.

Solution steps	Calculations
Step 1: Write dissociation equation.	$\text{LiCl(s)} \rightarrow$
Step 2: Identify which ion undergoes hydrolysis.	No hydrolysis
Step 3: State the answer.	Therefore

Worked example 2

Calculate the pH of 0.10 mol dm^{-3} of ammonium chloride (K_a of $\text{NH}_4^+ = 5.6 \times 10^{-10}$).

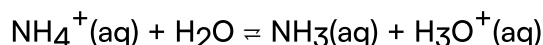
Dissociation of salt:



Therefore:

$$[\text{NH}_4^+] = 0.1 \text{ mol dm}^{-3}$$

Hydrolysis of ammonium, not chloride:



$$K_a = \frac{[\text{NH}_3][\text{H}_3\text{O}^+]}{[\text{NH}_4^+]}$$

As per previous K_a calculations, at equilibrium:

$$[\text{NH}_3] = [\text{H}_3\text{O}^+] = x$$

$$[\text{NH}_4^+] = 0.10 - x = 0.10$$

(remember the approximation since K_a is small)

$$K_a = 5.6 \times 10^{-10}$$

$$= \frac{x^2}{0.10}$$

Solving for x:

$$x = 7.5 \times 10^{-6}$$

Therefore:

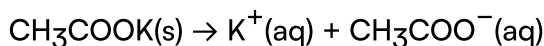
$$[\text{H}_3\text{O}^+] = 7.5 \times 10^{-6} \text{ mol dm}^{-3}$$

$$\begin{aligned} \text{pH} &= -\log [\text{H}_3\text{O}^+] \\ &= 5.13 \\ &= 5.1 \text{ (2 s.f.)} \end{aligned}$$

Worked example 3

Calculate the pH of 0.10 mol dm^{-3} of potassium ethanoate (K_b of $\text{CH}_3\text{COO}^- = 5.6 \times 10^{-10}$).

Dissociation of salt:



Therefore:

$$[\text{CH}_3\text{COO}^-] = 0.1 \text{ mol dm}^{-3}$$

Hydrolysis of ethanoate, not chloride:



$$K_b = \frac{[\text{CH}_3\text{COOH}] [\text{OH}^-]}{[\text{CH}_3\text{COO}^-]}$$

As per previous K_b calculations, at equilibrium:

$$[\text{CH}_3\text{COOH}] = [\text{OH}^-] = x$$

$$[\text{CH}_3\text{COO}^-] = 0.10 - x = 0.10$$

(remember the approximation since K_b is small)

$$K_a = 5.6 \times 10^{-10}$$

$$= \frac{x^2}{0.10}$$

Solving for x:

$$x = 7.5 \times 10^{-6}$$

Therefore:

$$[\text{OH}^-] = 7.5 \times 10^{-6} \text{ mol dm}^{-3}$$

$$[\text{OH}^-] \times [\text{H}_3\text{O}^+] = 1.0 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$$

Therefore:

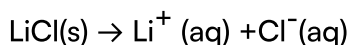
$$[\text{H}_3\text{O}^+] = 1.3 \times 10^{-9} \text{ mol dm}^{-3}$$

$$\begin{aligned}\text{pH} &= -\log [\text{H}_3\text{O}^+] \\ &= 8.87 \\ &= 8.9 \text{ (2 s.f.)}\end{aligned}$$

Worked example 4

Calculate the pH of 0.10 mol dm^{-3} lithium chloride.

Dissociation of salt:



No hydrolysis possible. Ions are neutral.

Therefore:

$$\text{pH} = 7$$

Activity

- **IB learner profile attribute:** Knowledgeable:
- **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 20 minutes
- **Activity type:** Individual activity

Instructions

1. For the salts listed in the table, state the cation, the anion, the acidic/basic behaviour of the ions, and conclude about the acidic/basic behaviour of the salt.
2. Research the pH of these salts. Are your predictions correct?

Salt	Cation	Acidic? Basic? Neutral?	Anion	Acidic? Basic? Neutral?
Ammonium nitrate				
Lithium fluoride				
Sodium carbonate				
Potassium ethanoate				
Lithium hydrogencarbonate				

pH curves of strong and weak monoprotic acids and bases (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to interpret the general shapes of pH curves for all four combinations of strong and weak acids and bases.

In [section R3.1.8 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ph-curves-for-strong-acid-base-neutralisation-reactions-id-46057/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ph-curves-for-strong-acid-base-neutralisation-reactions-id-46057/), you learned that titration is one of the most widely used techniques in quantitative analysis to determine concentrations of acids and bases. It is used in the food, pharmaceutical and cosmetic industries as an important analytical technique. Titration curves can give a lot of information regarding the acid or base. But do titration curves for all combinations of strong and weak acids and bases share the same characteristics?



Figure 1. Titration is used in the food, pharmaceutical and cosmetic industries.

Credit: Nico Tekol, Getty Images

The curve of a strong acid–strong base titration was examined in [section R3.1.8](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ph-curves-for-strong-acid-base-neutralisation-reactions-id-46057/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ph-curves-for-strong-acid-base-neutralisation-reactions-id-46057/>). In this section we will look at the curves of titrations of strong acids–weak bases, weak acids–strong bases, and weak acids–weak bases.

The pH curve for a weak acid—strong base titration

Figure 2 shows the pH curve for the addition of a strong base to a weak acid. Can you think of similarities and differences with the titration curve of a strong acid–strong base titration?

The curve starts at pH 3, which is the pH of the weak acid used to produce this pH curve. At first, the addition of the strong base results in the pH staying fairly constant; this is known as the buffer region. Then there is a sharp increase in pH from 7 to 11 and the curve flattens out at pH 13, the pH of a strong base. While the pH at the equivalence point for the titration of a strong acid with a strong base was 7, here it is not: the pH at the equivalence point is greater than 7. This is because of the basic salt that is produced from the neutralisation reaction, for example here, potassium ethanoate, CH_3COOK .

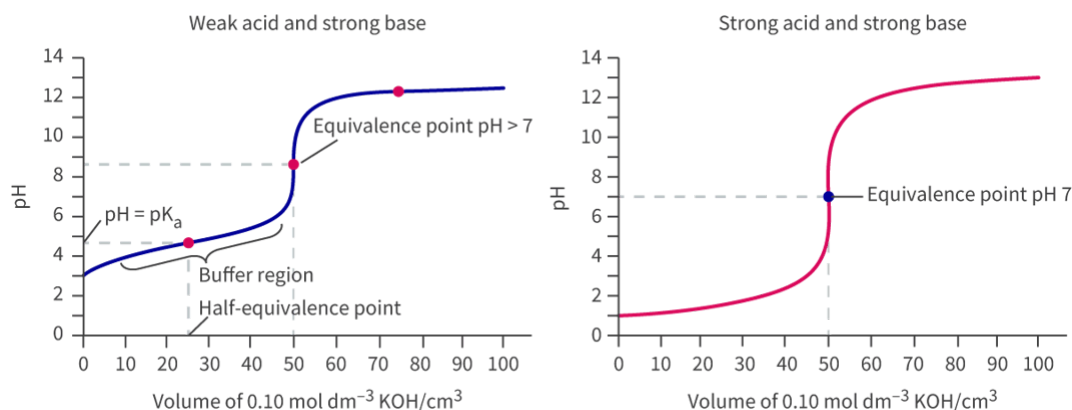


Figure 2. Titration curves for a weak acid and a strong base and strong acid and a strong base.

 More information for figure 2

This image contains two titration curves, each depicting a different chemical reaction.

1. **Left Graph (Weak Acid and Strong Base):**
2. **X-axis:** Represents the volume of $0.10 \text{ mol dm}^{-3} \text{ KOH/cm}^3$, ranging from 0 to 100 cm^3 .
3. **Y-axis:** Represents pH, ranging from 0 to 14.

4. **Curve Description:** Initially starts at a pH of around 3. As KOH is added, the pH remains relatively constant due to the buffer region, indicated between approximately 20 cm³ and 40 cm³. This region is followed by a sharp increase in pH between 40 cm³ and 50 cm³, indicating the equivalence point with a pH greater than 7. The curve flattens at a pH of about 13.
5. **Special Labels:** Marks include "pH = pK_a," "Buffer region," "Half-equivalence point," and "Equivalence point pH > 7."
6. **Right Graph (Strong Acid and Strong Base):**
7. **X-axis:** Similar setup as the first graph with volume of KOH/cm³ from 0 to 100 cm³.
8. **Y-axis:** Similar scale as the first graph with pH from 0 to 14.
9. **Curve Description:** Starts at an acidic pH around 1, increases more significantly after 40 cm³, with the equivalence point at 50 cm³ marked by a pH of 7. Beyond this point, the graph shows a less steep increase, leveling off at a pH near 13.
10. **Special Labels:** Includes "Equivalence point pH 7."

The graphs illustrate the differing behaviors of weak and strong acids when titrated with strong bases, highlighting the buffer region and the varying pH at the equivalence point.

[Generated by AI]

As shown on the first graph of **Figure 2**, there is a region in which pH remains relatively constant labelled as the buffer region. The buffer region is where the solution contains HA (the weak acid) and a significant amount of A⁻ (the conjugate base of the weak acid). At this point, the solution acts as a buffer solution, meaning that it has the ability to resist a change in pH on the addition of a small amount of acid or base. We call this the half-equivalence point.

The half-equivalence point can be found on the x-axis at the point which is half the volume added at the equivalence point. On the pH curve for a weak acid and a strong base in **Figure 2**, this corresponds to the 25 cm³ point. At the half-equivalence point, exactly half the acid has reacted with the base and been neutralised, with the other half remaining unreacted. The weak acid is in excess by half the amount added, the strong base has reacted completely. The conjugate base of the weak acid has been produced from the neutralisation in exactly the same amount as the reacted weak acid. The weak acid and conjugate base are found in solution in equal amounts. Therefore, the properties of the solution at this point make it an ideal buffer solution, which explains why the pH remains fairly constant when a small amount of base is added to the acid.

At the half-equivalence point, the pH is equal to the pK_a of the acid. The pH at the half-equivalence point can be read directly from the graph (draw a line up to the curve, then across to the y-axis), and this can be used to deduce the pK_a of the acid. At this point, the K_a = [H⁺], therefore, the pK_a = pH.

Study skills

When presented with a pH curve, make sure that you are able to find the pH at the equivalence point.

Practical skills

Tool 1: Experimental techniques — Applying techniques

When collecting data to generate a pH curve, when should smaller volumes of titrant be added between each measurement?

pH curve for a weak base—strong acid titration

Based on what you have learned so far, can you try plotting the titration curve for the titration for adding a strong acid to a weak base, for example hydrochloric acid and ammonia?

What do you think the most important features are? Consider the following prompts:

- What do you think the initial pH would be before any acid is added?
- Do you think pH will increase or decrease as more acid is added?
- What do you think the final pH will be after an excess amount of acid is added?
- What do you think the pH at the equivalence point will be, 7, above 7 or below 7?

Use **Interactive 1** to investigate the titration curve for a weak base/strong acid titration.

Interactive 1. The titration curve for a weak base—strong acid titration.

The curve starts at pH 11–12, the pH of the weak base used to produce this pH curve, and then decreases gradually in the buffer region. In this part of the graph, the solution resists changes in pH as the strong acid is added. There is then a steep drop in pH when 25 cm³ of acid has been added. The equivalence point is at a pH of less than 7, because of the acidic salt (NH₄Cl) that is produced.

In the buffer region, at the half-equivalence point, the pH of the conjugate acid of the weak base is equal to pK_a , or $pOH = pK_b$ for the weak base.

Study skills

You should be able to interpret pH curves for all combinations. The interpretation should include:

- intercept with the pH axis
- equivalence point
- buffer region
- points where $pH = pK_a$ or $pOH = pK_b$

pH curve for a weak base—weak acid titration

Based on what you have learned so far, can you try plotting the titration curve for adding a weak acid to a weak base? For example, ethanoic acid and ammonia.

What do you think the most important features are? What will be different from all graphs seen previously?

Use **Interactive 2** to investigate the titration curve for a weak base–weak acid titration.

Interactive 2. The Titration Curve for a Weak Base—Weak Acid Titration.

Interactive 2 shows the pH curve for the addition of a weak acid to a weak base. The curve starts at around pH 12, the pH of the weak base used to produce this pH curve, and then decreases to the pH of the added weak acid. The equivalence point is at a pH of 7, because of the neutral salt produced.

The starting pH is not high, the ending pH is not low, and there is a very small change in pH. The equivalence point is, therefore, difficult to determine and the graph is not particularly useful.

Study skills

It is possible to carry out these titrations the other way round — with the alkali in the flask, and adding the acid from the burette, or with the acid in the flask and adding the alkali from the burette. The curves obtained in this case are simply reflections of those we have considered here.

Practical skills

Tool 2: Technology — Applying technology to collect data

Tool 3: Mathematics - Graphing, Sketch graphs, with labelled but unscaled axis to qualitatively describe trends, Construct and interpret graphs for raw and processed data, Draw lines or curves of

best fit

In the lab, you can perform a titration of a weak acid with a strong base to determine the pK_a of the weak acid experimentally. You can collect pH using a data-logger, then graph the data and draw lines of best fit. How can you use the graph to determine the pK_a ?

Use **Interactive 3** to review the features of different titration curves.

Specify if the titration curve is for a weak acid–weak base, strong acid–strong base, weak acid–strong base, or strong acid–weak base combination.

pH at the equivalence point is 8.5.

Turn

Card 1 of 4



Interactive 3. Review the features of different titration curves.

Activity

- **IB learner profile attribute:** Thinker
- **Approaches to learning:**
 - Thinking skills — Being curious about the natural world; Applying key ideas and facts in new contexts
 - Social skills — Working collaboratively to achieve a common goal
- **Time required to complete activity:** 20 minutes

- **Activity type:** Pair activity

Instructions

1. Open the titration curve generator  (https://iwant2study.org/lookangejss/chemistry/2025_04_14/) which has titrations for six combinations: strong acid—strong base, strong base—strong acid, weak acid—strong base, weak base—strong acid, strong base—weak base, and strong acid—weak base.
2. Set-up each titration by choosing the concentration of acid and base. You can also choose either monoprotic or diprotic and the volume of solution in the conical flask.
3. Choose an indicator that you think will cover the pH range containing the equivalence point.
4. Run each titration and study the pH curves produced.
5. Try each of the different combinations of acids and bases and compare the pH curves produced.

Acid–base indicators (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to:

- (R3.1.14) Construct equilibria expressions to show why the colour of an indicator changes with pH.
- (R3.1.15) Identify an appropriate indicator for a titration from the identity of the salt and the pH range of the indicator.

In section R3.1.8 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ph-curves-for-strong-acid-base-neutralisation-reactions-id-46057/>), the use of indicators such as phenolphthalein to indicate the equivalence point was introduced. Did you know that many acid–base indicators can be found in nature or in household materials? Natural indicators are in foods like red cabbage and berries, or in flowers. The varying acidity of the soil can change the colour of some flowers from blue or pink to purple since the flowers contain some type of substance that changes colour with pH. Beautiful displays can be seen in garden in which the soil acidity is changed. Can we make use of such changes to understand more about acid–base reactions?



Figure 1. The different colours of the same flower.

Credit: Amana Images Inc, Getty Images

Watch **Video 1** to find out more about natural indicators.

How to make a home pH indicator from blueberries (si...

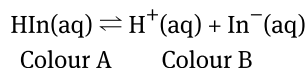


Video 1. How to use a natural indicator to indicate pH.

🕒 More information for video 1

How acid–base indicators work

An acid–base indicator is a weak acid or a weak base in which the dissociated and undissociated forms have different colours. Like all weak acids and weak bases, they exist in an equilibrium between the two forms, which are represented as HIn (the undissociated form) and In^- (the dissociated form) as seen in section R3.1.9–10 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ion-dissociation-constants-of-weak-acids-and-bases-hl-id-46058/>). The equation for the dissociation of HIn can be represented as:



where HIn and In^- have different colours (colour A and colour B). As this is an equilibrium reaction, the position of equilibrium can be changed by adding an acid (H^+ ions) or a base (OH^- ions) [section R2.3.4](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/le-chateliers-principle-id-46069/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/le-chateliers-principle-id-46069/>). Depending on the concentration of H^+ ions in solution (or OH^- ions), the solution takes one colour or the other.

Study skills

The generalised formula HIn(aq) can be used to represent the undissociated form of an indicator.

Figure 2 shows the undissociated and dissociated forms of phenolphthalein, which are colourless and pink respectively.

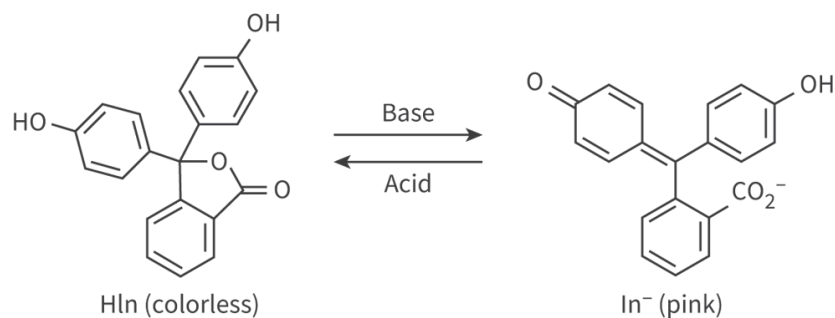


Figure 2. Phenolphthalein in its undissociated (HIn) and dissociated (In^-) forms.

 More information for figure 2

The image depicts the chemical equilibrium between the undissociated (HIn) and dissociated (In^-) forms of phenolphthalein. On the left, the structure of the undissociated form, HIn, is shown as a large molecule with three benzene rings connected by oxygen and hydroxyl groups. This form is noted to be colorless. On the right, the dissociated form, In^- , is depicted as a similar molecular structure but includes an additional carboxylate group (CO_2^-), which breaks the symmetry. This form is indicated to be pink. Arrows labeled 'Base' and 'Acid' indicate the reversible reaction between the two forms based on pH changes.

[Generated by AI]

Applying Le Chatelier's principle [section R2.3.4](#)

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/le-chateliers-principle-id-46069/>), we can predict how this equilibrium will respond to a change in pH.

- Increasing $[\text{H}^+]$ by adding an acid will shift the equilibrium to the left, so phenolphthalein appears colourless.
- Increasing $[\text{OH}^-]$ by adding a base will shift the equilibrium to the right, so phenolphthalein appears pink.

To summarise, the addition of an acid shifts the position of equilibrium to the side of the undissociated form (HIn), while the addition of a base shifts the equilibrium to the side of the dissociated form (In^-).

Indicators change colour when their $\text{pH} = \text{p}K_a$.

At equilibrium:

$$K_a = \frac{[\text{H}^+][\text{In}^-]}{[\text{HIn}]}$$

Indicators change colour when $[\text{In}^-] = [\text{HIn}]$ and therefore $K_a = [\text{H}^+]$.

Table 1 shows a list of indicators, their pH ranges for the colour change and the corresponding $\text{p}K_a$ values.

Table 1. A list of indicators, with their pH ranges and $\text{p}K_a$ values.

Indicator	pK_a	pH range	Colour in acid	Colour in
Methyl orange	3.7	3.1–4.4	red	yellow
Bromophenol blue	4.2	3.0–4.6	yellow	blue
Bromocresol green	4.7	3.8–5.4	yellow	blue
Methyl red	5.1	4.4–6.2	red	yellow
Bromothymol blue	7.0	6.0–7.6	yellow	blue
Phenol red	7.9	6.8–8.4	yellow	red
Phenolphthalein	9.6	8.3–10.0	colourless	pink

Study skills

Some acid–base indicators with pH ranges and pK_a values are listed in [section 18 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/acid-base-indicators-id-45120/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/acid-base-indicators-id-45120/) of the DP chemistry data booklet.

Note that indicators do not necessarily change colour around pH 7. Most indicators are weak acids with their own value of pK_a – they change colour over a pH range that centres on that value. For example, the change in colour of bromothymol blue, known as its end-point, happens in the range pH 6.0–7.6, which is within the approximate range of the $pK_a \pm 1.0$.

Study skills

Take care not to confuse the equivalence point and the end-point.

- The end-point is the pH when the indicator changes colour.
- The equivalence point is when acid and base have reacted in stoichiometrically equivalent amounts.

A suitable indicator for a titration is one for which these two points coincide, so the indicator changes colour near to the equivalence point. The indicator is also undergoing its own titration simultaneously to the titration that is the focus of the experiment. The indicator reaches its buffer region for the pH range and when the pH is equal the pK_a , both colours of the indicator are seen – that is the ideal choice for the titration because the colour transition is the one that should show.

Figure 3 shows how the colour change of bromothymol blue is capable of detecting the equivalence point in the titration of a strong alkali with a strong acid. The yellow and blue are both shown in a very brief window, appearing green – the green colour is the colour that should show for the titration – that is where the pH at equivalence

is matching the pK_a of the indicator. The key factor here is that the colour change takes place over a small addition of acid, coinciding with the vertical inflection in the pH curve.

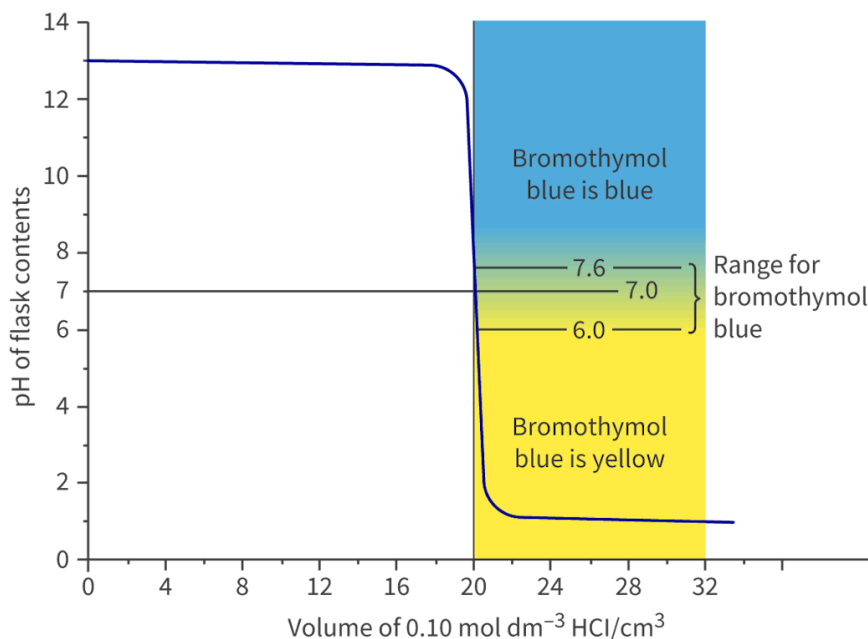


Figure 3. A strong alkali–strong acid titration curve showing the colour change for bromothymol blue.

 More information for figure 3

The graph illustrates a titration curve of a strong alkali reacting with a strong acid, using bromothymol blue as an indicator. The x-axis represents the volume of $0.10 \text{ mol/dm}^3 \text{ HCl}$ (in cm^3), ranging from 0 to 32, while the y-axis represents the pH of the flask contents, ranging from 0 to 14. The curve remains nearly horizontal with a high pH on the left side before sharply dropping near the equivalence point. The vertical drop signifies a rapid pH change due to the addition of acid. The area where the indicator changes color is highlighted: bromothymol blue is yellow below pH 6.0 and blue above pH 7.6, with the equivalence range marking a transition around pH 7.0. The pH change over a small volume of acid addition is critical, aligning with the steep inflection of the curve.

[Generated by AI]

Figure 4 shows the relationship between the pH range over which phenolphthalein changes colour and the pH curve for a strong alkali–strong acid titration. This shows that even though phenolphthalein changes colour at approximately pH 9.6, it is still an appropriate indicator for this titration. This is because the range over which the colour changes falls within the vertical region of the curve.

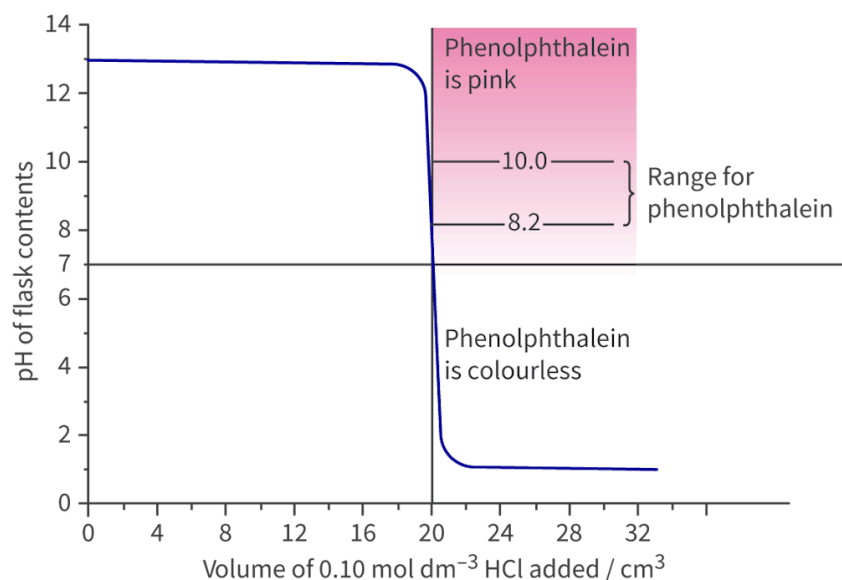


Figure 4. A strong alkali—strong acid titration curve using phenolphthalein.

 More information for figure 4

The graph shows a titration curve for a strong alkali–strong acid with phenolphthalein as an indicator. The X-axis represents the volume of 0.10 mol dm^{-3} HCl added in cm^3 , ranging from 0 to 34. The Y-axis represents the pH of the flask contents, ranging from 0 to 14.

The curve starts at a high pH level around 14 and remains level for the initial part until approximately 20 cm^3 of HCl is added. At this point, the curve sharply declines, indicating the equivalence point and a significant pH change within a small volume, common in strong acid–strong base reactions.

Phenolphthalein's color change range, between 8.2 and 10.0 pH, is marked on the graph. In this range, phenolphthalein appears pink. Below 8.2 pH, it is colorless, indicating the endpoint of the titration.

[Generated by AI]

Figure 3 and **Figure 4** show that both phenolphthalein and bromothymol blue would be appropriate indicators for titrations between a strong acid and a strong alkali. In fact, the size of the pH change around the equivalence point in such titrations is sufficiently great that several different indicators could be used.

Looking at the titration curves for other acid–base combinations, it is possible to make a judgement as to which indicators are suitable to use for a given titration.

Figure 5 shows that while phenolphthalein is an appropriate indicator for a titration of a strong base with a weak acid, methyl orange is not. This is because the pH range over which it changes colour (its end-point) does not fall within the vertical region of the curve.

Remember that at the equivalence point, the acid and base have reacted completely, and the salt has been formed. The pH at the equivalence point is the pH of the salt. Therefore, to find a suitable indicator for a titration:

1. Determine which acid and base are reacting.
2. Find the pH of the salt at the equivalence point.

3. Choose an indicator with an end point in the range of the equivalence point or close enough to it, i.e., where its pK_a is equal or close enough to the pH at the equivalence point.

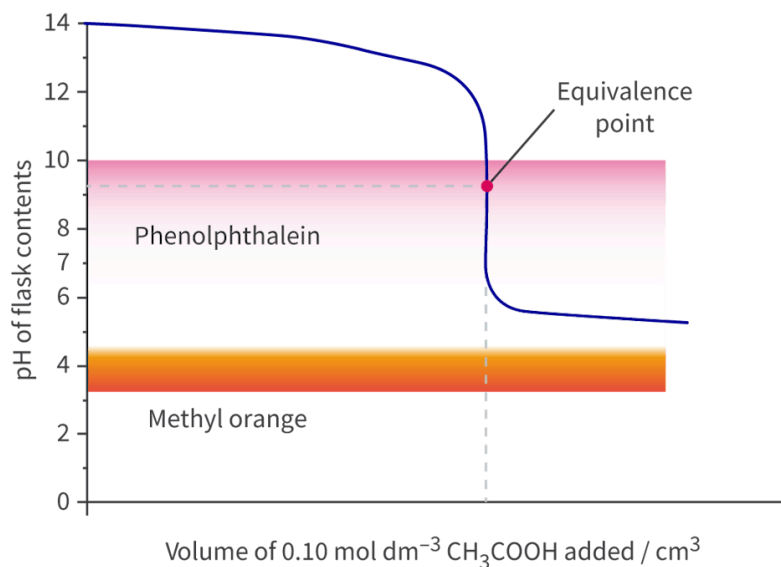


Figure 5. A strong alkali—weak acid titration curve using phenolphthalein and methyl orange.

 More information for figure 5

The image displays a titration curve with a blue line representing the change in pH as a strong alkali (base) is titrated with a weak acid. The X-axis shows the volume of 0.10 mol dm⁻³ CH₃COOH added in cm³, while the Y-axis shows the pH of the flask contents ranging from 0 to 14. The curve exhibits a steep rise near the equivalence point.

The graph highlights two indicators: "Phenolphthalein" and "Methyl orange," each marked by shaded areas on the pH scale. "Phenolphthalein" ranges from around pH 8 to pH 10, showing where it is effective as a pH indicator, while "Methyl orange" covers pH 3 to pH 4.

The "Equivalence point" is labeled on the curve, marked by a red dot where the slope is steepest, indicating the point at which stoichiometrically equivalent amounts of acid and base have reacted. This point is important for choosing the correct indicator to determine the transition colour close to the equivalence point, as noted in the additional context about detecting pale pink with phenolphthalein.

[Generated by AI]

Remember that you want the indicator to go through the colour change near the equivalence point. You want to see the transition colour at the equivalence point. For phenolphthalein, you will be able to detect the equivalence point if the end point or the transition colour (pale pink) coincides with the equivalence point.

Use **Interactive 1** to match different acid/base titrations with a suitable indicator.

Using the information on indicators in the data booklet:

What is a suitable indicator for a strong acid–strong base titration?

Turn

Card 1 of 4



Interactive 1. Review choice of suitable indicator.

Making connections

What are some of the similarities and differences between indicators used in acid–base titrations and in redox titrations? (see [section R3.2](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43486/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43486/>)).

Activity

- **IB learner profile attribute:** Thinker
- **Approaches to learning:**
 - Thinking skills — Being curious about the natural world; Applying key ideas and facts in new contexts
 - Social skills —Working collaboratively to achieve a common goal
- **Time required to complete activity:** 20 minutes
- **Activity type:** Pair activity

Instructions

Go on a hunt!

1. Find at least four chemical indicators at home or used for industrial purposes.
2. Include their names and their uses.
3. Find their pK_a and pH range values and include the equilibrium equation using correct formulas of the acid and ionised form of the indicator.

Conclude with justification for which titration (strong acid—strong base, strong acid—weak base or weak acid—strong base) they could be used in order to indicate the equivalence point.

Buffer solutions (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to:

- (R3.1.16) Describe the composition of acidic and basic buffers and explain their actions.
- (R3.1.17) Solve problems involving the composition and pH of a buffer solution, using the equilibrium constant.

We have seen solutions that are acidic, alkaline and neutral and the effect on pH when these solutions undergo titration. There are many natural systems at equilibrium where the pH of the solution needs to remain constant to protect the organism or ecosystem.

One example is your blood. The pH of your blood remains fairly constant, even when acidic or alkaline substances travel through your bloodstream. If the pH in your blood changes, it would have an effect on your health, leading to respiratory difficulties, liver failure, even brain damage.

When a solution resists changes in pH when acidic or alkaline substances are added, it is known as a buffer solution or simply a buffer. Buffer solutions have many uses in chemistry – they are utilised whenever it is necessary to have a controlled pH. Buffers are used to calibrate pH meter and are also an important ingredient in many consumer products, such as shampoos and dyes.

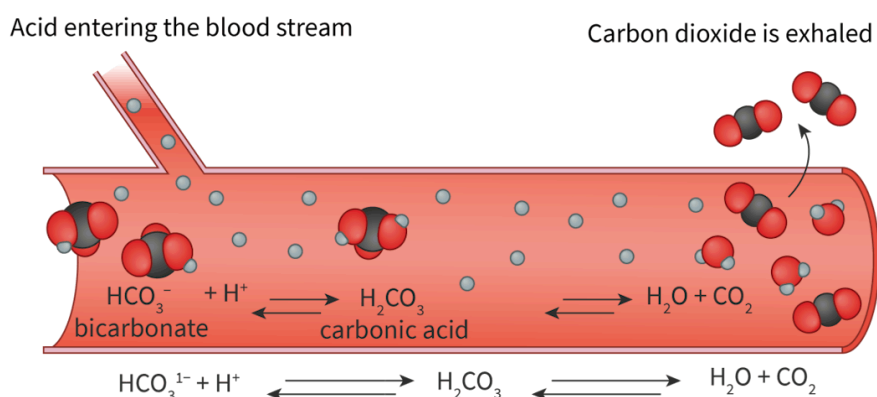


Figure 1. The buffer in blood.

More information for figure 1

The image is a diagram illustrating the buffer system in blood. It depicts a blood vessel with molecules and chemical reactions labeled within it. On the left side, labeled "Acid entering the blood stream," a reaction is shown between hydrogen ions (H^+) and bicarbonate ions (HCO_3^-) to form carbonic acid (H_2CO_3). This is further detailed with an arrow pointing left showing the equation $\text{HCO}_3^- + \text{H}^+ \leftrightarrow \text{H}_2\text{CO}_3$. To the right, the conversion of carbonic acid

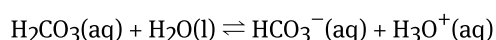
to carbon dioxide (CO₂) and water (H₂O) is illustrated with arrows and the equation $\text{H}_2\text{CO}_3 \leftrightarrow \text{H}_2\text{O} + \text{CO}_2$. On the far right, labeled "Carbon dioxide is exhaled," CO₂ molecules are depicted exiting the blood vessel. These interactions demonstrate how blood buffers changes in pH via reversible reactions.

[Generated by AI]

Buffer solutions

How exactly does a buffer solution work to resist a change in pH? Recall that a system at equilibrium that undergoes a stress, such as the addition of small amounts of acid or base, will respond by shifting the position of equilibrium (see [section R2.3.4 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/le-chateliers-principle-id-46069/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/le-chateliers-principle-id-46069/)).

The buffer system in your blood uses carbonic acid and carbonate ions to maintain the pH of blood at an average 7.40.



When a small amount of acid is added to a solution, the pH would be expected to decrease due to the addition of H⁺ ions to the solution. When small amounts of acid are added to a system at equilibrium, the system can respond by shifting the position of equilibrium to minimise the stress. According to Le Chatelier's principle, the increase in concentration of the H⁺ ions introduced by the acid, will cause the position of equilibrium to shift to the left (to favour the reactants), consuming the added H⁺ ions and therefore restoring the pH to 7.40.



When a small amount of base is added to a solution, the pH would be expected to increase due to the addition of OH⁻ ions to the solution. When small amounts of base are added a buffer, the system can respond by shifting the position of equilibrium. According to Le Chatelier's principle, the increase in concentration of the OH⁻ ions introduced by the base, will react with the H⁺ ions in solution, causing the concentration of H⁺ to decrease. This will cause the position of equilibrium to shift to the right (to favour the products), replacing the lost H⁺ ions and therefore restoring the pH to 7.40.



Figure 2 shows the effect of adding a small amount of acid or alkali to pure water. Note the significant change in pH that occurs from pH 7 to either pH 4 (for the addition of an acid) or to pH 10 (for the addition of an alkali). If the same amount of acid or base were added to a buffer solution, we could expect to see a much smaller change in the pH.

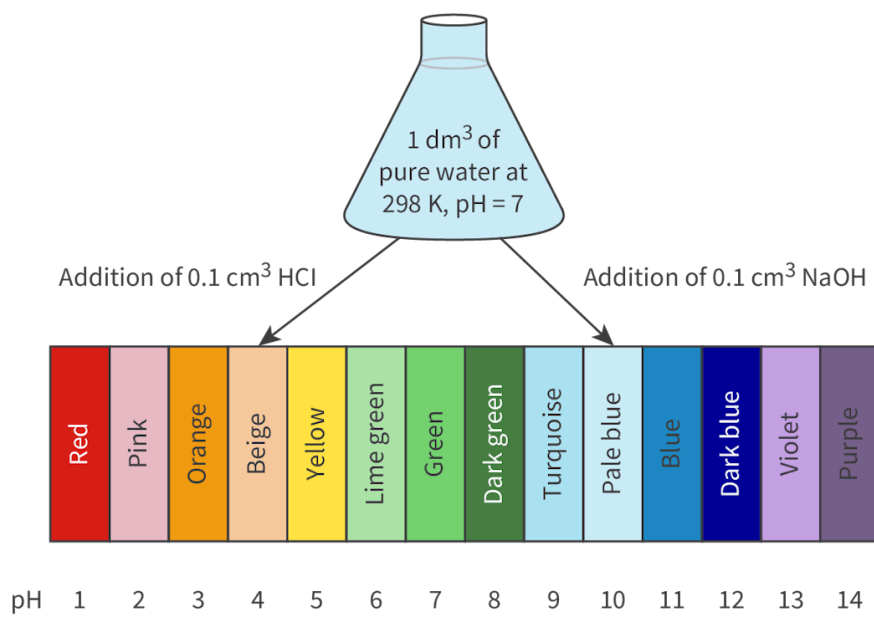


Figure 2. The change in pH on addition of a small amount of acid or alkali to pure water.

More information for figure 2

The diagram illustrates the change in pH of water when a small amount of acid (HCl) or alkali (NaOH) is added. At the center of the image is a conical flask labeled '1 dm³ of pure water at 298 K, pH = 7'. Two arrows extend from this flask: one pointing left, labeled 'Addition of 0.1 cm³ HCl', and the other pointing right, labeled 'Addition of 0.1 cm³ NaOH'. Below the arrows, a rectangular color scale displays pH levels with corresponding colors, ranging from pH 1 to 14. The colors range from red (pH 1) progressing through pink, orange, beige, yellow, lime green, green, dark green, turquoise, pale blue, blue, dark blue, violet, to purple (pH 14). The left arrow (addition of HCl) indicates a decrease in pH towards 4 (beige), while the right arrow (addition of NaOH) indicates an increase towards pH 10 (pale blue).

[Generated by AI]

Watch **Video 1** to see what would happen if an acid or base was added to a buffer solution.

Buffer Demonstration 2 0 for Avid



Video 1. How a buffer keeps pH constant until a certain point.

Adding small amounts of acid or base to a buffer solution maintains the pH relatively constant. This comparative effect on pH is shown in **Figure 3**.

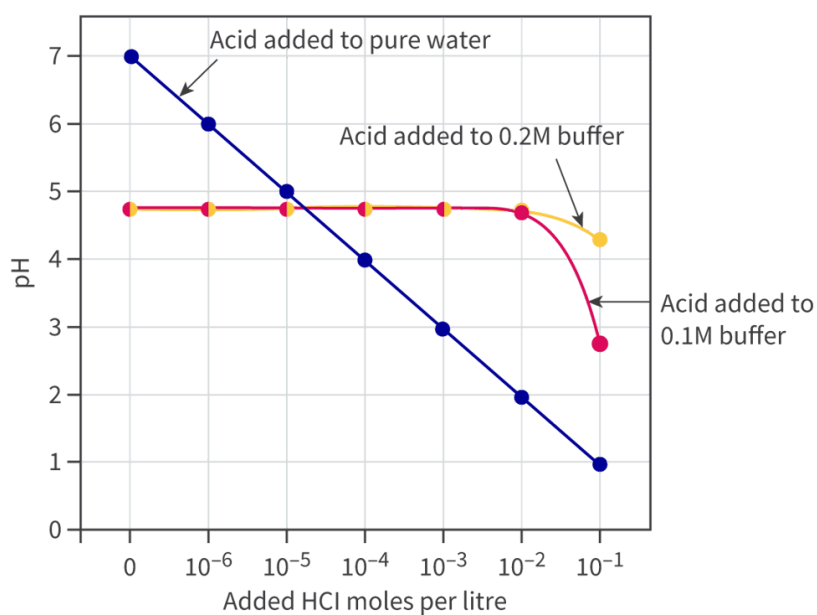


Figure 3. The change in pH on addition of a small amount of acid or alkali to pure water and to a buffer solution.

More information for figure 3

The graph illustrates how the pH changes when small amounts of hydrochloric acid (HCl) are added to pure water and two types of buffer solutions. The X-axis of the graph is labeled "Added HCl moles per litre" and presents a logarithmic scale from 0 to 10^{-1} . The Y-axis is labeled "pH" and ranges from 0 to 7.

Three different curves are shown on the graph: 1. A blue line showing the acid added to pure water. It starts at the upper left corner with a pH of 7 and decreases steadily as more acid is added. 2. A yellow line indicating the data for a 0.2M buffer solution. The pH remains constant around 5 even with an increase in HCl up to 10^{-3} moles per litre, then drops slightly. 3. A red line

showing the 0.1M buffer solution. Initially, the pH remains around 5 but begins to decline gradually after 10^{-3} moles per litre, dropping significantly past 10^{-2} moles per litre.

Overall, the graph demonstrates the buffering capacity, showing that adding acid to buffer solutions maintains a relatively constant pH compared to pure water.

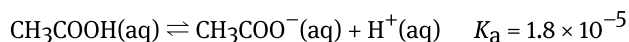
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Theory of Knowledge

Buffer solutions were originally used to simulate biological systems. Experiments involving live cell cultures typically require a buffer medium to maintain pH for optimal biological processes. How does the concept of equilibrium in buffer solutions relate to the nature of scientific knowledge?

Composition of buffers

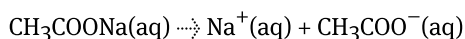
Recall that weak acids and bases ionise partially, existing as a system at equilibrium, whereas strong acids and bases ionise completely (see [section R3.1.6a](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/strong-and-weak-acids-and-bases-id-44703/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/strong-and-weak-acids-and-bases-id-44703/>)). Therefore, buffers require a weak acid or base as the primary component. For example, ethanoic acid is a weak acid that can be used to make a buffer solution. The dissociation of ethanoic acid and its K_a value are shown here.



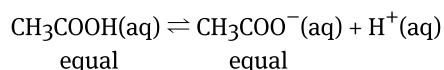
In order to make a buffer solution that can respond to stress, the system at equilibrium must contain a relatively equal amount of both the weak acid and the conjugate base ($\text{CH}_3\text{COOH} / \text{CH}_3\text{COO}^-$) – otherwise, the response to the stress would not be very effective.

However, a weak acid or base does not ionise enough to provide a sufficient concentration of the conjugate base, since its K_a is very low. Therefore, the conjugate base must be added to the weak acid to provide a sufficient concentration.

How can the conjugate base (CH_3COO^-) be added to the solution? Recall that salts can have acid–base activity (see [section R3.1.12](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ph-of-salt-solutions-hl-id-46060/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ph-of-salt-solutions-hl-id-46060/>)). The ethanoate ion can be added to the solution by adding a salt that contains the ethanoate ion as the anion, but care must be taken to select a cation that will not provide any acid–base activity to the buffer solution. An example of a suitable salt is sodium ethanoate, CH_3COONa , which dissociates according to the following equation:

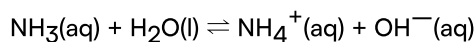


The combination of equal amounts of ethanoic acid and sodium ethanoate will provide relatively equal amounts of the weak acid and its conjugate base at equilibrium, allowing for effective buffer action.



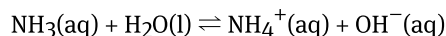
Using your knowledge of how to make a buffer solution starting with a weak acid, how would you make a buffer starting with a weak base, such as ammonia, NH_3 ?

The weak base ammonia, NH_3 , must also have its conjugate acid present in a relatively equal concentration for the buffer to respond effectively to stress.



The conjugate acid of ammonia is ammonium, NH_4^+ . Ammonium ions can be added by using a salt that contains ammonium ions as the cation, but the anion must not have any acid–base activity. One example for a suitable salt is ammonium chloride, NH_4Cl .

For this buffer system, the equilibrium equation is:



If small amounts of acid are added, the H^+ ions from the acid will neutralise the OH^- ions in solution, decreasing their concentration and causing the equilibrium to shift to the right to replace them and restore the pH. Conversely, if small amounts of base are added, the concentration of OH^- ions in solution increases, causing the equilibrium to shift to the left to consume them, which then restores the pH.

There is another way to make a buffer solution. Can you recall another time that you have seen the term buffer before. In [section R3.1.13](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ph-curves-of-strong-and-weak-monoprotic-acids-and-bases-hl-id-46061/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ph-curves-of-strong-and-weak-monoprotic-acids-and-bases-hl-id-46061/>) a weak acid (or base) was titrated with a strong base (or acid) before reaching the equivalence point. There is an area on the titration curve known as the buffer region. This is where the partially titrated acid(or base) contains a sufficient amount of its conjugate base (or acid), such that a buffer solution is made (**Figure 4**).

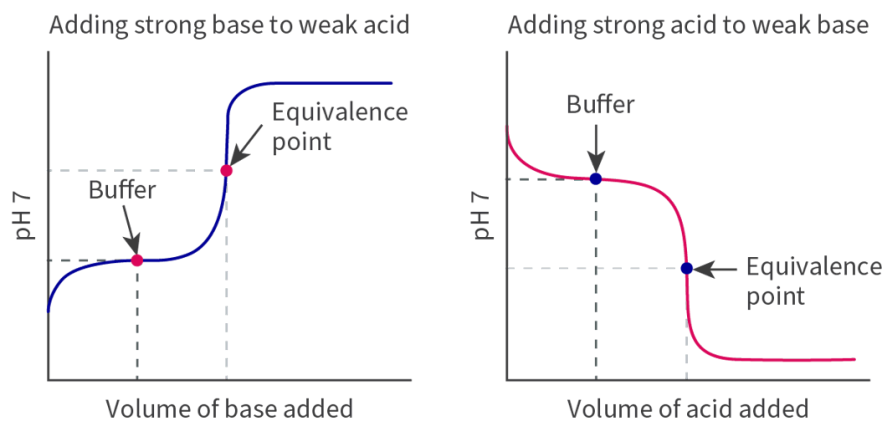


Figure 4. A buffer can be made by partially neutralising a weak acid or base.

 More information for figure 4

The image contains two graphs illustrating titration curves.

On the left, the graph titled "Adding strong base to weak acid" displays a blue titration curve. The x-axis is labeled "Volume of base added," and the y-axis is labeled "pH." The curve starts at a low pH, increases gradually, then sharply rises, and finally levels off after the equivalence point. The buffer region is marked and identified before the steep rise begins. The equivalence point is indicated.

On the right, the graph titled "Adding strong acid to weak base" shows a red titration curve. The x-axis is labeled "Volume of acid added," and the y-axis is again labeled "pH." This curve starts at a high pH, decreases gradually, then sharply drops, and eventually levels off beyond the equivalence point. The buffer region is marked and identified prior to the sharp decline. The equivalence point is also indicated in this graph.

[Generated by AI]

Worked example 1

Which of the following mixtures forms a buffer solution with a pH below 7?

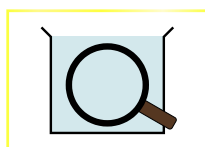
1. 0.0200 dm^3 of 0.10 mol dm^{-3} carbonic acid is mixed with 0.0100 dm^3 of 0.10 mol dm^{-3} sodium hydroxide.
2. 0.0200 dm^3 of 0.10 mol dm^{-3} carbonic acid is mixed with 0.0200 dm^3 of 0.10 mol dm^{-3} sodium hydroxide.
3. 0.0100 dm^3 of 0.10 mol dm^{-3} carbonic acid is mixed with 0.0200 dm^3 of 0.10 mol dm^{-3} sodium hydroxide.

1. will result in a buffer solution with a pH of less than 7 since the moles of strong base are half of the number of moles of weak acid, resulting in partially neutralising the weak acid.
2. will result in the acid is being completely neutralised since the moles of strong base are equal to the moles of weak acid. This solution will not act as a buffer solution.
3. will result in an excess of strong base as the moles of strong base are greater than the moles of weak acid. The pH is more than 7 as too much base is added. This solution will not act as a buffer solution.

These combinations show the mixing of a strong base and a weak acid. For a buffer to be formed, we saw that enough of the strong base should be added to half neutralise the weak acid. The only combination that matches this is 1.

Study skills

If you are having a hard time visualising these processes, you can access the simulation below and see this at a molecular level:



Interactive 1. Explore pH: Interactive Acid-Base Solutions Simulation.

 More information for interactive 1

An interactive simulation demonstrates acid-base solutions through two panels labeled Intro and My Solution, each of which zooms in on its corresponding illustration when selected.

In the intro panel, a panel on the right allows users to select a solution from the following options:

- Water (H_2O)
- Strong Acid (HA)
- Weak Acid (HA)
- Strong Base (MOH)
- Weak Base (B)

A beaker in the center contains 1 liter of a selected solution, with a magnifying glass revealing the particles inside. Users can immerse a pH meter into the solution to find its respective pH values.

Another panel on the right allows users to select different viewing options, such as displaying particles, a bar graph for equilibrium concentration, and hide views.

Below the beaker is the reaction of the respective solution. There are also options to switch the pH meter to a pH scale and a circuit for checking electrical conductivity.

The reaction and equilibrium concentrations of the reactants and products are as follows:

For water:

- Reaction: $2 \text{H}_2\text{O} \rightleftharpoons \text{H}_3\text{O}^+ + \text{OH}^-$
- Equilibrium concentrations of H_2O , H_3O^+ , and OH^- are 55.6, 1.0×10^{-7} , and 1.0×10^{-7} moles per liter, respectively.
- The pH is 7.00.
- Poor conductor of electricity.

For Strong Acid (HA):

- Reaction: $\text{HA} + \text{H}_2\text{O} \rightarrow \text{A}^- + \text{H}_3\text{O}^+$
- Equilibrium concentrations of HA, H_2O , A^- , and H_3O^+ are negligible, 55.6, 1.0×10^{-2} , and 1.0×10^{-2} moles per liter, respectively.
- The pH is 2.00.
- Strong conductor of electricity.

For Weak Acid (HA):

- Reaction: $\text{HA} + \text{H}_2\text{O} \rightleftharpoons \text{A}^- + \text{H}_3\text{O}^+$
- Equilibrium concentrations of HA, H_2O , A^- , and H_3O^+ are 1.0×10^{-2} , 55.6, 3.2×10^{-5} , and 3.2×10^{-5} moles per liter, respectively.
- The pH is 4.50.
- Weak conductor of electricity.

For Strong Base (MOH):

- Reaction: $M^+OH^- \rightarrow M^+ + OH^-$
- Equilibrium concentrations of M^+OH^- , M^+ , and OH^- are negligible, 1.0×10^{-2} , and 1.0×10^{-2} moles per liter, respectively.
- The pH is 12.00.
- Strong conductor of electricity.

For Weak Base (B):

- Reaction: $B + H_2O \rightleftharpoons BH^+ + OH^-$
- Equilibrium concentrations of B, H_2O , BH^+ , and OH^- are 1.0×10^{-2} , 55.6, 3.2×10^{-5} , and 3.2×10^{-5} moles per liter, respectively.
- The pH is 9.50.
- Strong conductor of electricity.

In the My Solution panel, the same phenomenon is exhibited. Users can check the pH of the weak and strong acids and bases by changing the initial concentration, ranging from 0.001 to 1 mole per liter. Users can also adjust a slider to change the solution strength from weaker to stronger.

This simulation allows users to understand that as the initial concentration increases from 0.001 to 1 mole per liter, the pH of the weak acid decreases, the pH of the weak base increases, the pH of the strong acid decreases, and the pH of the strong base increases.

Table 1 shows the four different ways to make a buffer:

Table 1. Ways to make a buffer.

	Begin with:	Then add:	Final pH of buffer solution
Method 1	Weak acid	Salt containing the conjugate base	pH < 7
Method 2	Weak base	Salt containing the conjugate acid	pH > 7
Method 3	Weak acid	A strong base to half neutralise the weak acid	pH < 7
Method 4	Weak base	A strong acid to half neutralise the weak base	pH > 7

Making connections

Can you think about the role of buffers and explain why buffer solutions must be composed of weak acid or base conjugate systems, not of strong acids or bases? What can weak acids/bases do that strong acids/bases cannot? (see [subtopic R2.3](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43482/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43482/>)).

Because a buffer solution consists of an equilibrium system between acid and conjugate base, when adding a small amount of strong acid or base, Le Châtelier's principle can be applied to explain how the pH remains constant. How does Le Châtelier's principle enable us to interpret the behaviour of indicators and buffer solutions? (see [subtopic R2.3](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43482/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43482/>)).

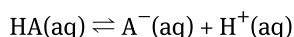
The pH of a buffer

When preparing to make a buffer solution, it is important to select the appropriate components so that the buffer will have the desired pH required for the experiment. The pH of a buffer is determined by:

- the pK_a or pK_b of the weak acid (or base);
- the relative ratio of the concentrations of weak acid (or weak base) and the conjugate base (or conjugate acid)

Calculating the pH of an acidic buffer

We saw that an acidic buffer solution consists of a weak acid HA and its conjugate base A^- in similar mole quantities. The concentration of the weak acid is represented as HA and the concentration of the conjugate base is represented as A^- . When the two solutions are added together, there will be a dilution that occurs and must be corrected.



Using the equilibrium expression and K_a (or pK_a) of the weak acid,

$$K_a = \frac{[H^+][A^-]}{[HA]}$$

When the $[H^+]$ is determined, the pH is easily calculated using the formula,

$$pH = -\log[H^+]$$

Worked example 2

A student is making a sample of simulated gastric juice in the stomach by preparing a buffer for an experiment.

0.0200 dm^3 of 0.10 mol dm^{-3} carbonic acid (H_2CO_3) is added to 0.0100 dm^3 of 0.10 mol dm^{-3} sodium hydrogen carbonate ($NaHCO_3$).

Calculate the pH of the buffer solution.

K_a of carbonic acid is 4.47×10^{-7} .

First, the new concentrations must be calculated for dilution in the new total volume:

$$0.0200 \text{ dm}^3 + 0.0100 \text{ dm}^3 = 0.0300 \text{ dm}^3 \text{ total volume}$$

The concentration of carbonic acid is:

$$\frac{0.0200 \times 0.10}{0.0300} = 0.0667 \text{ mol dm}^{-3} \text{ H}_2\text{CO}_3$$

The concentration of sodium hydrogen carbonate is:

$$\frac{0.0100 \times 0.10}{0.0300} = 0.0333 \text{ mol dm}^{-3} \text{ NaHCO}_3$$

Since NaHCO_3 dissolves in the aqueous solution, it is used as the initial concentration of the conjugate base before the equilibrium is reached.

Using an ICE table, fill in the information that is known. The initial concentrations of H_2CO_3 and NaHCO_3 were calculated and before the acid dissociates, $[\text{H}^+] = 0$.

The reaction then shifts to the right side to favour its formation of the products until equilibrium is established.

Step/Variable	Equilibrium E
Let x be the amount of acid dissociated to reach equilibrium.	$\text{H}_2\text{CO}_3(\text{aq})$
Initial concentration (mol dm^{-3})	0.0667
Change in concentration (mol dm^{-3})	$-x$
Equilibrium concentration (mol dm^{-3})	$0.0667 - x$

Substituting into the K_a equation gives:

$$\begin{aligned}
 K_a &= \frac{[\text{H}^+][\text{HCO}_3^-]}{[\text{H}_2\text{CO}_3]} \\
 &= \frac{x(0.0333 + x)}{0.0667 - x}
 \end{aligned}$$

Since the K_a is very small, and the acid is weak, it dissociates very little, and therefore x is a very small number and is negligible when being added or subtracted from a much larger number. Therefore:

$$\begin{aligned}
 0.0333 + x &= 0.0333 \text{ and} \\
 0.0667 - x &= 0.0667
 \end{aligned}$$

Solving for x gives:

$$\begin{aligned}
 x &= \frac{0.0667}{0.0333} \times 4.47 \times 10^{-7} \\
 &= 8.95 \times 10^{-7}
 \end{aligned}$$

Therefore:

$$[\text{H}^+] = 8.95 \times 10^{-7} \text{ mol dm}^{-3}$$

$$\begin{aligned} \text{pH} &= -\log[\text{H}^+] \\ &= -\log 8.95 \times 10^{-7} \\ &= 6.05 \end{aligned}$$

The pH of this buffer solution is reasonable, given that it is made using a weak acid. The pH would be expected to be below 7, but not close to 1. Always consider the result of your calculations to see that the pH value makes sense.

Also note the significant figures. Since the concentration of carbonic acid is given to 2 significant figures, the final result will have 2 digits to the right of the decimal place (for pH values the digit to the left of the decimal place has no effect on the number of significant figures).

Calculating the pH of a basic buffer

Similarly, for a weak base B and its conjugate acid HB^+ :



Therefore:

$$\begin{aligned} K_b &= \frac{[\text{OH}^-][\text{HB}^+]}{[\text{B}]} \\ -\log_{10} K_b &= -\log_{10} \left(\frac{[\text{OH}^-][\text{HB}^+]}{[\text{B}]} \right) \\ -\log_{10} K_b &= -\log_{10} [\text{OH}^-] - \log_{10} \left(\frac{[\text{HB}^+]}{[\text{B}]} \right) \\ \text{p}K_b &= \text{pOH} - \log_{10} \left(\frac{[\text{HB}^+]}{[\text{B}]} \right) \end{aligned}$$

Rearrange to find pOH:

$$\begin{aligned} \text{pOH} &= \text{p}K_b + \log_{10} \left(\frac{[\text{HB}^+]}{[\text{B}]} \right) \\ \text{pH} &= 14 - \text{pOH} \end{aligned}$$

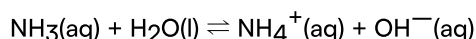
B is a weak base and does not dissociate completely and the initial concentration of the base B is the approximately the same as the final concentration and concentration of the conjugated acid HB^+ stays approximately the same hence, we can use the initial concentrations.

Worked example 3

Calculate the pH of the buffer solution prepared by combining 0.0667 mol ammonia and 0.0333 mol ammonium ions in a 1.00 dm^3 volumetric flask.

K_b of ammonia is 1.78×10^{-5}

The buffer system at equilibrium is:



Using an ICE table, fill in the information that is known. Since the total volume of the solution is 1.00 dm^3 , the number of moles is equal to the molar concentration. Also note that the initial concentration of OH^- will be 0 since the weak base has not dissociated yet. Equilibrium will then shift to the right, favouring the formation of the products.

Step/Variable	Equilibrium Equation		
Let x be the amount of base dissociated to reach equilibrium.	$\text{NH}_3(\text{aq})$	$\rightleftharpoons$	
Initial concentration (mol dm^{-3})	0.0667		
Change in concentration (mol dm^{-3})	$-x$		
Equilibrium concentration (mol dm^{-3})	$0.0667 - x$		

Substituting in the K_b equation gives:

$$\begin{aligned}
 K_b &= \frac{[\text{OH}^-][\text{NH}_4^+]}{[\text{NH}_3]} \\
 &= \frac{x(0.0333 + x)}{0.0667 - x}
 \end{aligned}$$

Since the K_b is very small, and the base is weak, it dissociates very little, and therefore x is a very small number and is negligible when being added or subtracted from a much larger number. Therefore:

$$0.0333 + x = 0.0333 \quad \text{and} \quad 0.0667 - x = 0.0667$$

Solving for x gives:

$$\begin{aligned}
 x &= \frac{0.0667}{0.0333} \times 1.78 \times 10^{-5} \\
 &= 3.56 \times 10^{-5}
 \end{aligned}$$

Therefore:

$$\begin{aligned}
 [\text{OH}^-] &= 3.56 \times 10^{-5} \text{ mol dm}^{-3} \\
 \text{pOH} &= -\log[\text{OH}^-] \\
 \text{pOH} &= -\log[3.56 \times 10^{-5} \text{ mol dm}^{-3}] \\
 &= 4.448 \\
 \text{pH} &= 14 - \text{pOH} \\
 \text{pH} &= 14 - 4.448 \\
 &= 9.55 \text{ (3 s.f.)}
 \end{aligned}$$

The pH of this buffer solution is therefore 9.552. This makes sense considering that the buffer is made with a weak base. The pH would be expected to be above 7, but not too close to 14. Always consider the result of your calculations to see that the pH value makes sense.

Also note the significant figures. Since the K_b is given to 3 significant figures, the final result of the logarithmic function in the pH calculation will have 3 digits to the right of the decimal place (for pH values the digit to the left of the decimal place has no effect on the number of significant figures).

Dilution of a buffer solution

Would diluting a buffer solution with distilled water affect its pH? Think about the components of a buffer solution. When adding water, the ratios of the weak acid: conjugate base (or weak base: conjugate acid) will remain the same, but the concentrations will be lower, so the response of the buffer will be reduced.

Consider the analogy of a balance in **Figure 5**. When the buffer solution is an equilibrium, the two sides are balanced. If a stress is applied to the system by adding small quantities of acid or base, one pan will drop. Le Châtelier's principle allows for balls to be shifted from the heavy side to the lighter side, restoring equilibrium.

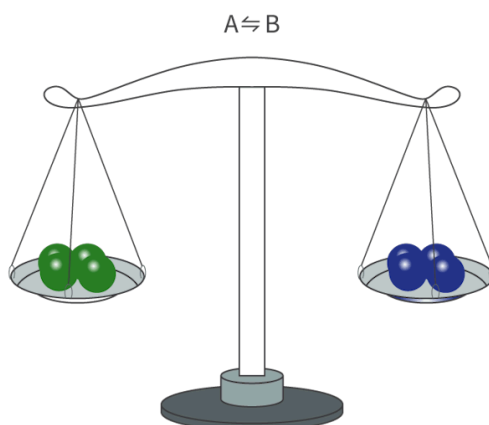


Figure 5. A buffer contains a balanced ratio of a weak acid (or base) and its conjugate base (or acid).

 More information for figure 5

The image is an illustration of a balance scale with two pans. On the left pan, there are two green spheres, while on the right pan, there are two blue spheres. Both sides are shown to be in equilibrium, representing a balanced state. This visual serves as a metaphor for a chemical buffer system, where the balance illustrates the equilibrium maintained between a weak acid and its conjugate base, or a weak base and its conjugate acid. The image conveys the concept that in a buffer system, similar to the balance scale, equilibrium can be maintained even when stress, in the form of a small amount of acid or base, is applied.

[Generated by AI]

Consider that there are two different balances. One balance has ten balls on each side and the other balance contains only four on each side. When a stress is applied, let's say two additional balls are added (**Figure 6**), which balance can respond to stress better?

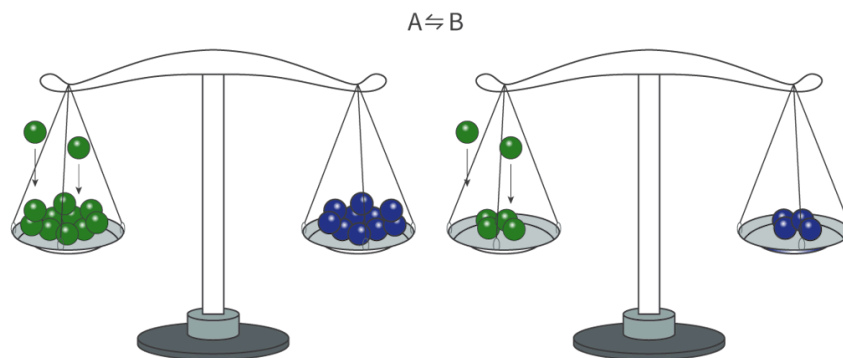


Figure 6. A more concentrated buffer can respond better to stress than a diluted buffer.

 More information for figure 6

The illustration depicts two balance scales, each with two sets of spheres representing buffer components. The balance on the left contains a larger number of green spheres on each side, with an additional two green spheres being added to one side, showing an imbalance. The balance on the right has fewer blue spheres on both sides with two additional spheres added on one side, creating a similar imbalance. Above the image, there's an equation symbol ' $A \rightleftharpoons B$ ', which represents chemical equilibrium. The image illustrates the concept of balanced versus stressed buffer systems, where a more concentrated buffer can better absorb additional reagents without losing equilibrium, compared to a diluted buffer.

[Generated by AI]

Both solutions are still buffers and will still respond by shifting the position of equilibrium to minimise the stress applied but remember that buffers can only resist pH changes when small amounts of acid or base are added. For a more concentrated buffer, the small amount of acid or base can be easily managed in relation to the higher concentrations of the buffer components, allowing equilibrium to be restored and the pH maintained. For a diluted buffer, there is not a high enough concentration of the buffer components to shift equilibrium sufficiently. Equilibrium cannot be restored and the pH will not be maintained.

Activity

- **IB learner profile attribute:** Inquirer
- **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 15 minutes
- **Activity type:** Pair activity

Instructions

Go on a hunt!

1. Research four or five buffer systems found in nature. Then do some research and fill in the blanks with the corresponding information in

the table below.

Buffer	pK_a or pK_b	Formula of the conjugate acid	F

1. Design an experiment that you could perform in the lab to prepare a buffer of pH5 and test its buffer capacity. What should you start with? Which materials would you need? Which calculations are needed? Which method can you use to test the buffer capacity?